Altitude-Targeting Rocket System

Leonardo Borchert, Shota Kakiuchi, Tyler Jones, Lean Albeshri

Mechanical and Aerospace Engineering, The George Washington University

MAE 4152W Capstone II, Final Technical Report

Submitted April 29, 2026

Team

Student Team Members (Authors)

Leonardo Borchert

Team Leader, Airframe, and Propulsion Expert

Tyler Jones

Mechanical Design, Manufacturing, and CFD

Shota Kakiuchi

Electronics and Software

Lean Albeshri

Recovery Avionics

Customers and Mentors

Professor Amir Barakati

Mentor

GW Rocket Team

Sponsor and Customer

Abstract

This capstone project presents a high power rocket system capable of targeting a pre-specified altitude through drag control. To do this, a rocket similar to those used at the International Rocket Engineering Competition was designed and modified with a deployable airbrake mechanism and flight computer implementing real-time sensor-based control.

The aerodynamic behavior of the rocket and brakes was predicted using computational fluid dynamics across several brake deployment states. In tandem, the brakes' structural integrity was validated via finite element analysis. The flight profile and propulsion system were modeled in OpenRocket and OpenMotor, respectively, giving predictions for trajectory and thrust profiles that guided the design of the rocket and its subsystems.

The onboard flight computer combines barometric and inertial data to run a control loop that determines the rocket's predicted trajectory and optimal airbrake deployment. The propulsion system was designed and manufactured by the team to ensure performance that represented a typical GW Rocket team flight. An active recovery system was included to ensure safe descent, so that the rocket could be reused.

The entire system was validated through a series of ground and flight tests, including parachute deployment, flight profile, aerodynamic performance, and brake deployment. Full-scale launches with and without active airbrake targeting were conducted to evaluate the altitude reduction potential of the airbrakes. These tests showed that the airbrakes could produce a 4.6% reduction in the rocket's apogee.

1. Introduction

1.1 Scope

The project involves the design of an airbrake system, software to control the system in flight, and the production of a rocket to fly and test the airbrakes. At its outset, the project focused solely on the development of the airbrakes, but it was quickly determined that a dedicated rocket would be necessary for proper flight testing. Additionally, the design of a capstone rocket allowed the team to optimize the airbrake placement for proper aerodynamic stability, and to fix the airbrake system's required weight early in the course of design.

The project was prescribed by the GW Rocket team. The motivation for an airbrake system on a rocket such as those used by the GW Rocket team is the precise control of the rocket's apogee, as prescribed by the International Rocket Engineering Competition, which the GW Rocket team competes in. The primary objective of the competition is to reach a specific predefined apogee with high accuracy, which an active airbrake is designed to achieve. This project's mission is to design such a system and then prove its function in flight.

1.2 Background

A. Aerodynamic Effects of Airbrakes

Deployable airbrakes reduce apogee by increasing drag during the coast phase of flight, between the engine burn and the peak of ascent, called the apogee.

Drag force follows the relation:

$$D = \frac{1}{2}\rho V^2 C_D A$$

where A is a constant reference area. For a rocket, this is typically the maximum cross-sectional area. The mathematical effect of brake deployment is an increase in C_d, which can be measured by experimentation or computational fluid dynamics simulation.

Studies of theoretical airbrake systems using simulation software report that acceptable stability margins are maintained when airbrakes are placed on the rocket's aft end and deploy symmetrically [1, 2, 3]. These studies also emphasize the importance of symmetric actuation. If the brakes do not deploy evenly, the center of pressure will become misaligned with the rocket's center axis, causing yawing torque and potential destabilization [2].

B. Brake Deployment Mechanisms

Rocket airbrake systems typically use one of three deployment pathways: extruding, folding/hinged, or rotating brakes, as shown in Figure 1.2.1 [1]. Mechanism selection is driven by tradeoffs in structural disruption, packaging constraints, achievable break area, and manufacturing complexity.

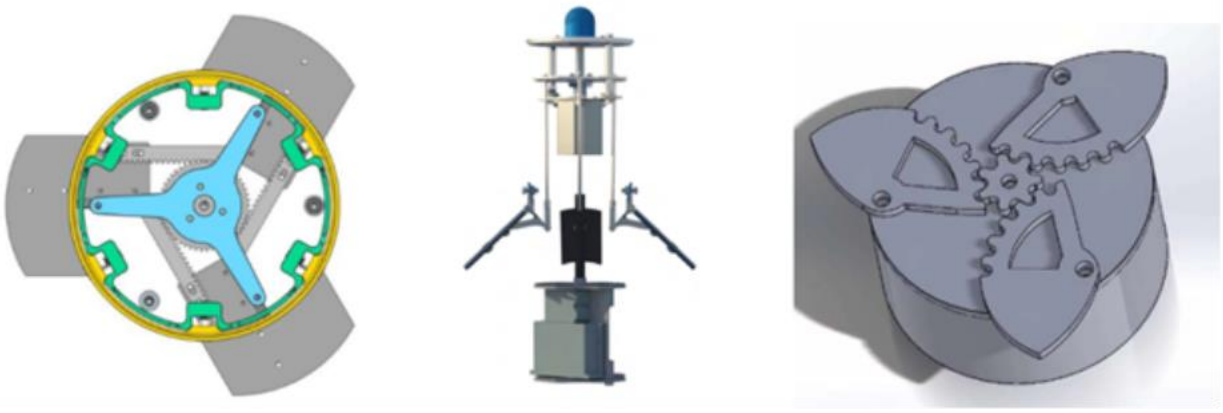


Figure 1.2.1: a) Extruding b) Folding c) Rotating

Considerations to be made when choosing between design pathways include the structural impact of the mechanism [1], the mechanism's weight and location relative to its centers of gravity and pressure (CG and CP), the total area and therefore total drag, the potential downstream disruption caused by the brakes' wake, and other case-specific factors like ease of manufacturing and cost. It must be ensured that the air brake is placed behind the CG so that stability cannot reach zero. Extruding brakes are by far the most common, and mechanisms for this design pathway can include rack/pinion sets as shown above as well as crank arms and spiral pin/slot joints.

It is desired that the brakes deploy as quickly as possible to their initial position upon engine shutdown, as well as precise control being needed to dial in the target apogee [1, 3].
Use of an Archimedes spiral, wherein $r = a\theta$ for any real number a, creates a linear relationship between the motor's and brakes' positions. This greatly simplifies the necessary software and makes control of the brakes at low displacement more precise than in other designs [4].

As long as a gentle spiral is used, this design should significantly reduce internal friction between the sides of the guiding plate and brake plates. This is because the force applied by the spiral slot is oriented mostly outward, whereas in a hinged plate design such as that shown to the left in Figure 1.2.2, the motor pushes the plates almost entirely to the side until they have been deployed a significant distance.

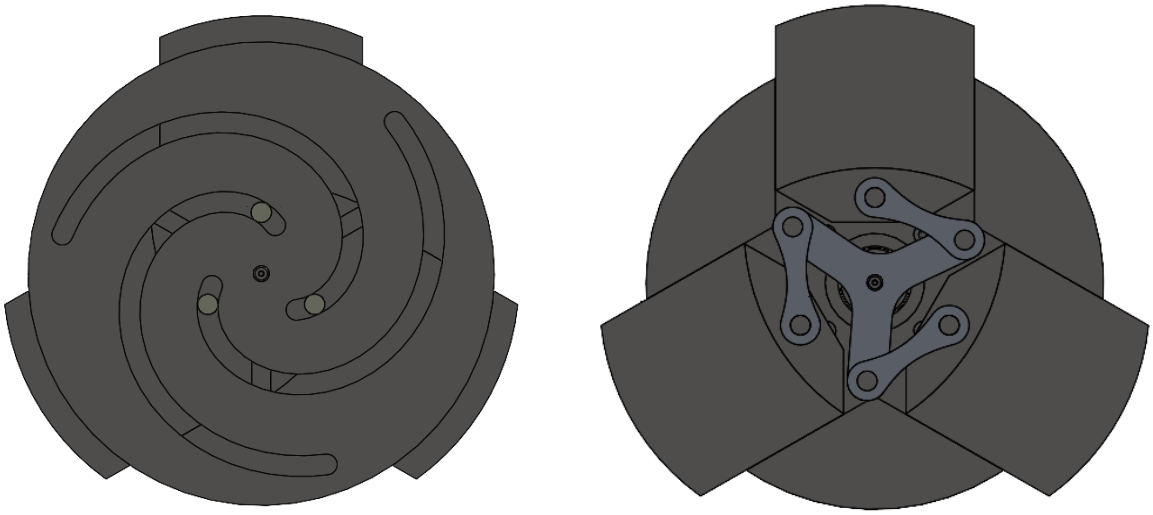


Figure 1.2.2: Spiral mechanism (left) versus hinge mechanism (right)

This becomes especially necessary during flight, when the brakes are under load and therefore pressed against the bottom of the guide plate, further increasing friction.

C. Software Modeling for Rocket and Airbrake Control

Closed-loop airbrake control depends on accurate software models that predict rocket behavior during the coast phase and translate those predictions into deployment commands.

There are three main parts of a controllable airbrake system; code, computation and modeling, and theoretical and mathematical principles. First, the RocketPy Python library provides a modular framework for simulating rocket flight dynamics and controlling airbrake deployment and timing [4]. Second, computational modeling allows for deeper understanding of the forces acting on the rocket and determining the effects [5, 6, 7]. Third, the fundamentals in controlling the airbrakes rely on theoretical and mathematical principles such as lift, drag, and acceleration.

The simulations result in accurate predictions only because they are based on underlying physics and aerodynamic principles. Understanding these relationships allows engineers to predict how changes in airbrake design will affect trajectory, enabling optimization for both stability and target apogee control.

D. Avionics Architectures for Rocket and Airbrake Control

Avionics are crucial to the project's development, enabling control over the airbrake system and the rocket's descent.

The Arbalest Rocket Goose 4 avionics system provides an example of an integrated design. It incorporates a Featherweight Blue Raven flight computer, previously used by the GW Rocket team and commercially-available, along with a custom printed circuit board (PCB) that manages power distribution, sensors, and deployment systems [8]. The system includes a Featherweight GPS tracker for redundancy, and communications are handled with the Adafruit LoRa RFM9X. This configuration demonstrates how the Blue Raven can be expanded with additional hardware to support robust flight, logging, payload, and recovery functions.

In contrast, the avionics configuration presented in [9] demonstrates a simpler architecture relative to the Arbalest system and is supported by NASA's F' software framework [9]. Another design shows how commercial components paired with a modular software framework can provide a reliable avionics package. The system was also developed for a smaller rocket, but it remains a potential option that could be adapted.

Beyond system design and control, physical constraints such as g-forces also impact avionics performance and selection. Broader studies on electronic durability in extreme conditions describe protective methods such as encapsulation, where epoxy fills protect circuit cavities, and underfill, where epoxy composites reinforce solder joints [10]. The main concern with g-forces is vibration, which could loosen or disconnect circuit components and fasteners. Properly securing all connections helps minimize this risk.

1.3 Functional Requirements

The customer requirements for this project were set by the performance of the GW Rocket team in the International Rocket Engineering Competition (IREC). In the competition, the team generally overshoots the desired apogee by several hundred feet. This established the need to develop an altitude-targeting system that functions as an independent subsystem while integrating smoothly with the rest of the rocket subsystems. The airbrakes must not interfere with other subsystems' functions or compromise the aerodynamic stability of the rocket during deployment, and it must fit within the specified space and mass constraints. To ensure the validity of these aspects of the design, and to test the performance of the airbrakes themselves, the capstone team also developed a dedicated rocket. In addition, the GW Rocket team's supplied funding along with the capstone class's budget determined the financial constraint that this project must stay within. Based on the needs of the GW Rocket team and the system outcomes they specified, the functional requirements for this system have been established as shown in Table 1.3.1.

Table 1.3.1 Functional Requirements

Functional Requirements	Category (Basic/Performance/Exciter)

Volume constraints: - Max diameter 5.875”, max height 14” - Airbrakes must extrude from the 2”-wide mid-coupler switchband	Basic
Airbrakes must reduce apogee by some degree	Basic
Airbrake system must weigh 15 lb maximum, 10 lb minimum (including supporting electronics i.e. electronics)	Basic
Airbrakes must withstand at least 22 lbf drag	Basic
Rocket must land safely for reuse	Basic
Total rocket cost should not exceed $5000	Performance
Rocket should reach a base apogee of at least 4500’	Performance
Manufacture of purpose-built rocket for airbrake testing	Performance
Project uses student researched-&-developed rocket engines in the manner of the GW Rocket engines	Exciter
Airbrake system reduces apogee by 7.5%	Exciter
Failure diagnostic and cinematic cameras	Exciter
Autonomous control system is flight-tested	Exciter

The customer’s direct expectations are expressed under the basic category, including altitude correction capability and compliance to the space and weight constraints of the coupler in which the airbrakes will be contained. The performance requirements set a benchmark for desirable targets such as achieving a minimum base apogee that is typically seen during a GW Rocket team subscale flight. The functional requirements also include aspects that are not required by the customer but will add value such as testing the airbrakes in flight, including cameras to observe the flight, and using engines that are very similar to those used by the GW rocket team.

The development of target values for the functional requirements was informed by simulations and background research. OpenRocket simulations determined the mass constraint for the airbrake system to ensure the rocket maintains a stability factor close to the ideal value of 2. Background research was conducted to estimate the drag force produced by the airbrakes and the theoretical apogee reduction percentage. OpenMotor simulations were used to estimate the thrust profile of the engine, which dictates the altitude the rocket can reach before airbrake deployment.

2. Design Description

2.1 Overview

Development of the rocket and airbrakes was split into subsystems similarly to how the GW Rocket team subdivides its work. These subsystems include the rocket airframe or aerostructures, the recovery system including hardware and electronics, the propulsion system, the airbrake mechanism and supporting structure, and the airbrake electronics and software.
The physical layout of these systems on the rocket is shown in Figure 2.1.1.

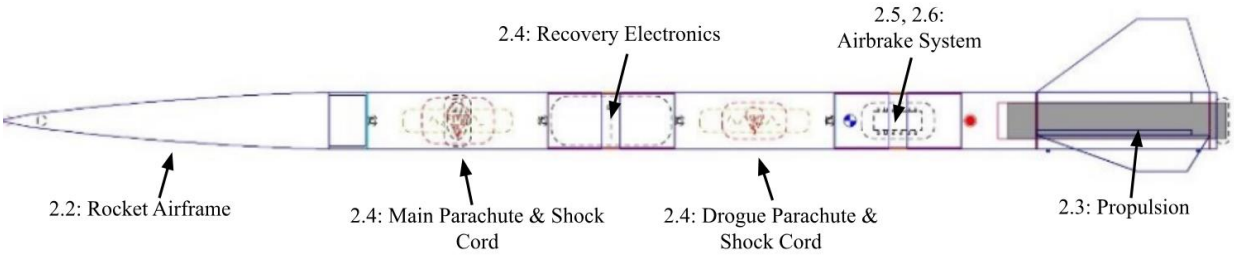


Figure 2.1.1: Rocket System Layout

Each aspect of the rocket is detailed in the following sections according to the numbering shown above.

2.2 Rocket Airframe & Layout

The rocket is designed to closely resemble those used by the GW Rocket team. It consists of 6.125” fiberglass body tube sections, 6” coupler sections, and the same commercially-available nose cone and fins used by the team. All the fiberglass components used to build the capstone rocket were taken from rockets used by the team in previous competitions and cut to size according to the specifications of the OpenRocket design, shown in Figure 2.2.1. From the nose cone to the fin can, left to right, body section lengths are 33.94,” 30,” 29.5,” and 34.” Additionally, there are three 2” switchbands forming the midsections of each coupler, bringing the total rocket height to 133.44.”

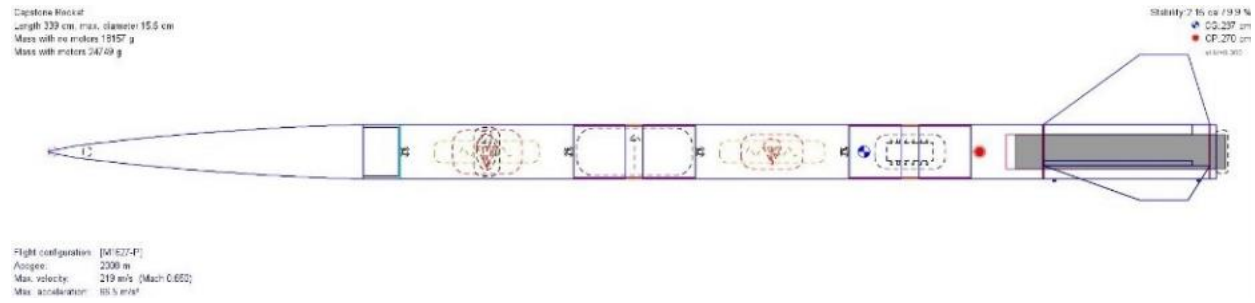


Figure 2.2.1: OpenRocket Simulation

The CAD model is shown in Figure 2.2.2.

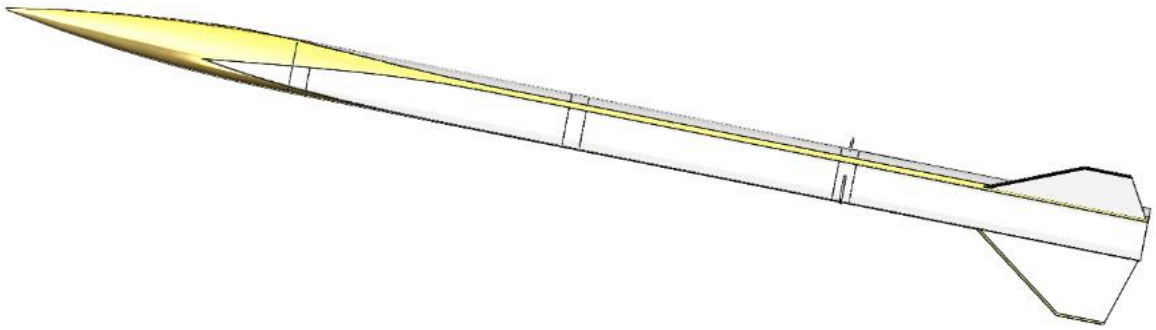


Figure 2.2.2: Rocket CAD Model

Finally, the complete airframe is shown prior to repainting in Figure 2.2.3.



Figure 2.2.3: Complete Airframe

After the rocket was built and painted, final weights were used to calculate the actual centers of gravity and pressure (CG, CP) before and after engine burn. The primary concern in the design of the rocket is the need to maintain proper stability, which is calculated as follows. [3]

$$SM = \frac{X_{CP} - X_{CG}}{D}$$

X_{CP} and X_{CG} are the distances of the centers of pressure and gravity from the tip of the nose cone, and D is the maximum diameter of the rocket body. Stability margin (SM) must be positive and should remain between 1.5 and 2.5.

There must be enough distance between the CG and CP that the rocket can self-straighten during ascent, but not so much as to cause weathercocking [11] or other adverse effects of overstability, where the fins have too much torque authority and correct the rocket's pitch too strongly. Adding to the complexity of the system, once the engine propellant is gone, the CG will shift much further forward, and when the brakes deploy, the CP will also shift forward, so these configurations must also be considered. This leads to the first design rule of an airbrake system; that it must always be placed behind the rocket's CG so that the stability cannot become

negative. Because of the potential for the brakes to destabilize the system, the capstone rocket was designed to be almost overstable with engine installed and no brakes deployed. Doing this necessitated that the airbrake coupler weigh 12 pounds in order to keep the rocket from being overstable. This is because the airbrake coupler is the aftmost place where additional weight can be placed.

With the weights finalized, a simulation was run for a launch at the Tripoli BattlePark site in Virginia. No brake deployment was simulated because the software cannot account for it (independently developed airbrake deployment simulations will be discussed in section 3.4). The result is shown in Figure 2.2.4.

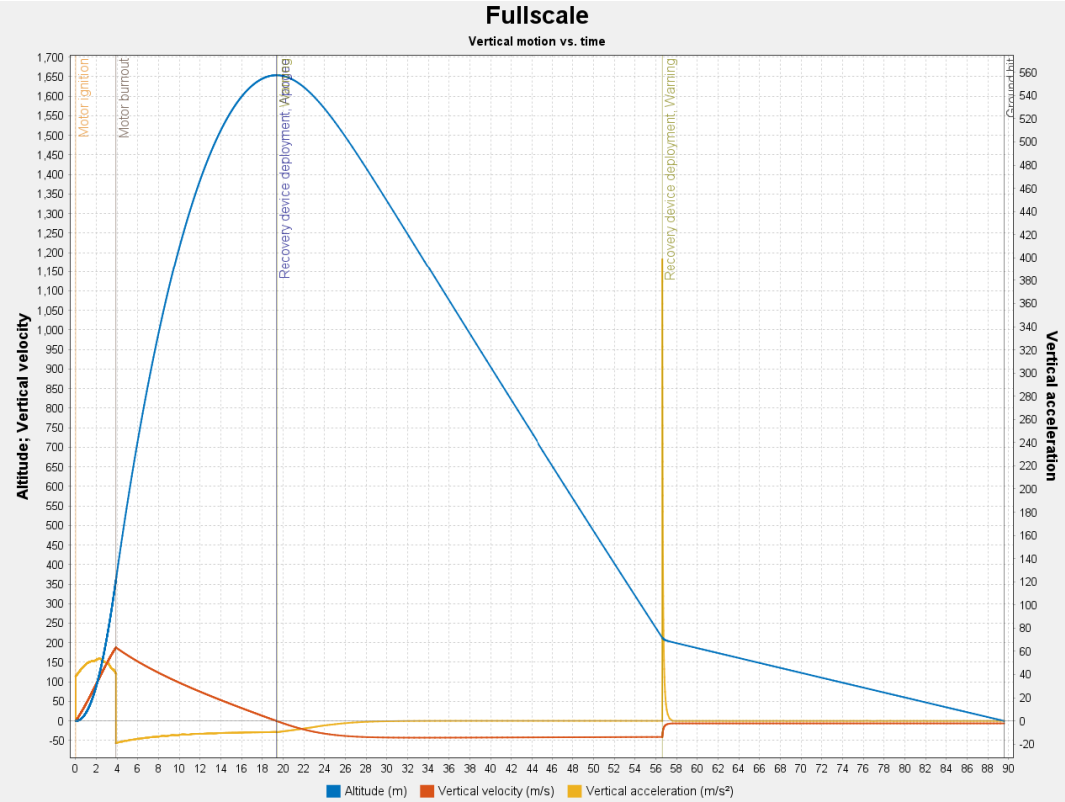


Figure 2.2.4: Launch simulation

The simulation estimated that the rocket would reach a maximum velocity of 191 m/s and a base apogee of 5567 ft. Additionally, with an estimated motor burn time of 4 seconds, it suggested that the time during which the airbrakes could be deployed is 15.6 seconds.

Table 2.2.1 Airframe: Status of Components

Item	Component name or description	To be fabricated or purchased	Detailed design completed or purchase specified	Design fabricated or

				purchase received
1	Body Tube	To be purchased & fabricated	Detailed design complete	Fabricated, assembled
2	Nose Cone	To be purchased	Purchase specified	Received
3	Fin-can	To be fabricated	Detailed design complete	Fabricated
4	Motor Tube	To be purchased	Purchase specified	Received
5	Hard & Shear Points	To be purchased	Purchase specified	Received
6	Couplers & Switch-band	To be purchased & fabricated	Detailed design completed	Fabricated, assembled
7	Paint	To be purchased	Purchase specified	Received
8	Bulkplates	To be manufactured	Detailed design completed	Fabricated, assembled

2.3 Propulsion Chemical Design

To ensure that the rocket's flight closely resembles those of the GW Rocket team, student researched-&-designed (SRAD) solid motors were manufactured for the rocket's flight tests, using the same mix as the GW Rocket engines. Because of this, testing was needed to validate the motors' effectiveness and safety.

The rocket's motor is an HTPB bimodal propellant. Four characterization tests have been performed to evaluate the pressure and thrust characteristics. Characterization tests are small motor burns that are conducted to validate the chemical mixture's effectiveness before large scale manufacturing is conducted. The propellant is composed of the following components:

Solids:

- Ammonium Perchlorate (AP, 400µm and 200µm)
- Aluminum (Al)
- Red Iron Oxide

Liquids:

- HTPB R-45HT
- IDP
- Tepanol - HX878
- AT Modified MDI

The red iron oxide (Fe_2O_3) functions as the ballistic modifier, added in small quantities to significantly increase how fast the propellant burns by catalyzing AP decomposition. AP

(NH_4ClO_4), the oxidizer, makes up the bulk of the propellant mass (75%). Aluminum is a high-energy inorganic fuel that burns with the oxidizer. It increases the combustion temperature and therefore exit velocity, which increases specific impulse. The IDP (Isodecyl Pelargonate) is the plasticizer. The HTPB and MDI increase the viscosity of the uncured propellant slurry, making it easier to mix and cast. MDI (Methylene diphenyl diisocyanate) serves as the curing agent and crosslinker, reacting with HTPB, hardening the liquid mix into a solid. In addition, the HTPB R-45HT (Hydroxyl-terminated polybutadiene) forms the solid matrix holding the solidified mix together. It also acts as the primary organic fuel, reacting with the AP's oxygen during combustion to form CO and CO_2. Finally, Tepanol consists of a mixture of chemicals and may reduce the end-mix viscosity and improve the mechanical properties of the cured propellant.

To understand how the motor will perform, an OpenMotor simulation was conducted. OpenMotor is a solid rocket motor simulation software that allows for single and multi-grain motors to be configured and tested. The motor grains are configured in the classical Bates configuration (cylindrical internal surface) as shown in Figure 2.3.1. Each grain is 5.75 inches tall, and this motor contains 3 grains.

	1	2	3
Port			
Mass	2.303 lb	2.303 lb	2.303 lb
Mass Flow (Port)	0.000 lb/s	0.000 lb/s	0.000 lb/s
Mass Flux (Port)	0.000 lb/(in^2*s)	0.000 lb/(in^2*s)	0.000 lb/(in^2*s)
Web	0.953 in	0.953 in	0.953 in

Figure 2.3.1: OpenMotor Bates Grain Configuration

The grains have a simulated burn duration of 3.83 seconds, and a thrust of 1730.28 N. The chamber pressure also lies in the 500 - 750 PSI range over the duration of the burn, which is ideal to reduce motor over-pressurization. The thrust and pressure profiles are detailed in Figure 2.3.2.

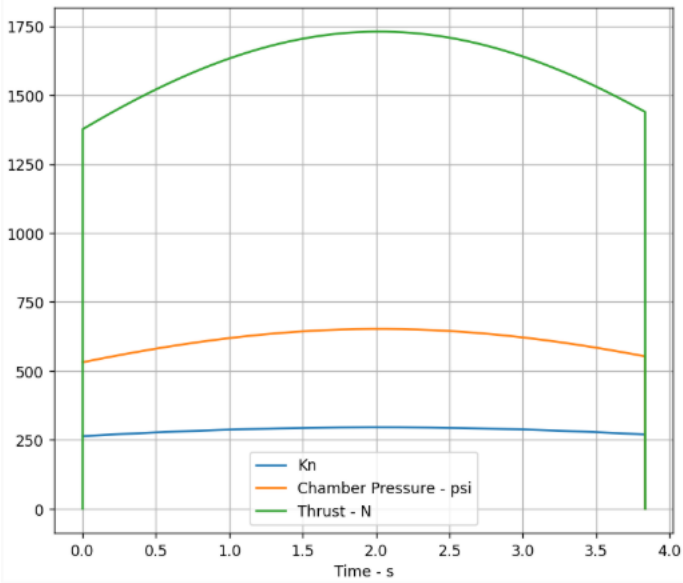


Figure 2.3.2: OpenMotor Bates Grain Simulation Results

Table 2.3.1 Propulsion: Status of Components

Item	Component name or description	To be fabricated or purchased	Detailed design completed or purchase specified	Design fabricated or purchase received
1	Propellant	To be fabricated	Detailed design complete	Fabricated
2	Phenolic Liner	To be purchased	Purchase specified	Purchase received
3	Nozzle	To be fabricated	Detailed design complete	Fabricated
4	Forward Closure	To be fabricated	Detailed design complete	Fabricated
5	Grease	To be purchased	Purchase specified	Purchase received
6	Motor Case	To be purchased	Purchase specified	Purchase received
7	Snap Rings	To be purchased	Purchase specified	Purchase received
8	O-rings	To be purchased	Purchase specified	Purchase received

2.4 Recovery Hardware & Avionics

The recovery system ensures the safe and controlled descent of the rocket. Recovery hardware includes an 18” drogue parachute and 40’ drogue shock cord, and a 96” main parachute and 30’ main shock cord. A ground test of the recovery system showed that 4 grams of black powder was able to separate the rocket’s stages and deploy the parachutes. Therefore, the primary charge wells were loaded with 4 grams of black powder per launch, and the secondary charge wells were loaded with 6 grams in case the 4 gram charge failed to separate the stages in flight.

All recovery electronics are housed within the recovery coupler, as shown in Figure 2.1.1, repeated here for clarity.

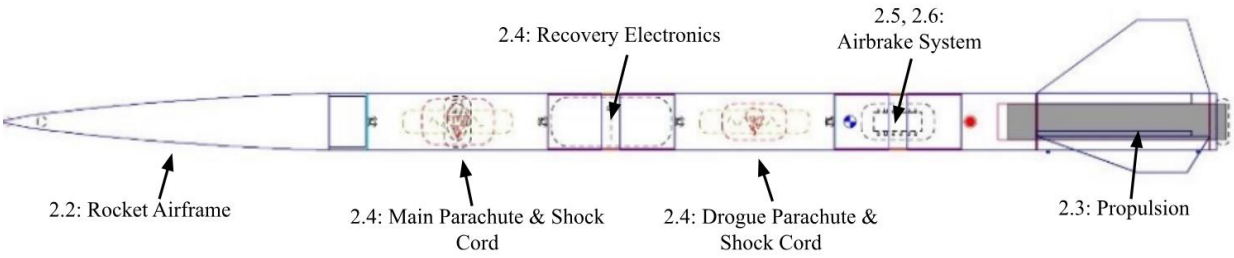


Figure 2.1.1: Rocket System Layout

The system includes two altimeters; the EasyMini as the primary system with the Missile Works RRC3+ as the secondary (backup). Initially, a BlueRaven altimeter was used with toggle switches to power on the altimeters. However, after the subscale flight (discussed in section 3.7) the Blue Raven was replaced with the RRC3+ and the toggle switches were replaced with screw switches such as those used by the GW Rocket team. This updated design provided redundancy which ensures reliable parachute deployment even in the event of an altimeter or charge well failure. In addition to the altimeters, the coupler included a Featherweight GPS tracker for locating the rocket after landing, and two onboard cameras oriented vertically and horizontally to capture video footage, as shown in Figure 2.4.1.



Figure 2.4.1: Onboard Camera Footage Screenshot

To ensure structural stability, three internal tiers were designed. The first tier is the avionics tier, which includes the altimeters and the GPS tracker. These components are mounted using screws and hex nuts, which were vibration-tested prior to the subscale launch. The second tier is the battery tier, which supplies power to both altimeters via two 9-volt alkaline batteries. This was designed to be identical to the GW Rocket team's recovery power supply. The third tier is the horizontal camera tier, which provides a stable and secure mount so that the camera remains fixed and properly oriented during flight. Lastly, a vertical camera mount was included to provide additional footage. The entire assembly is shown in Figure 2.4.2.

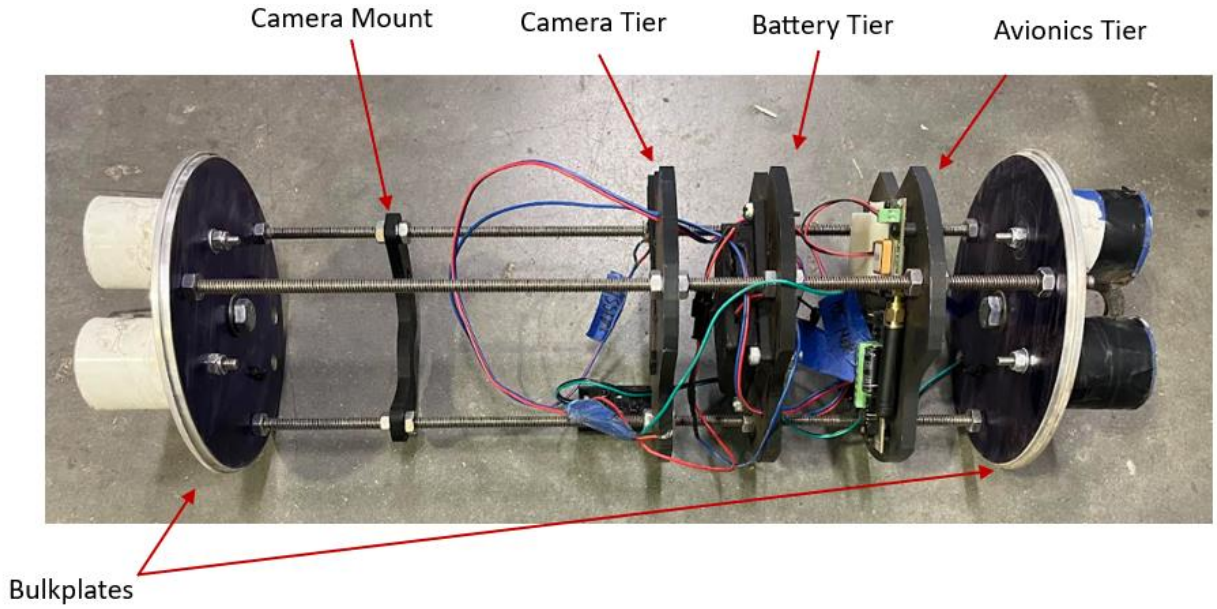


Figure 2.4.2: Recovery Tier Assembly

The tiers are connected using three threaded rods running the full length of the coupler. Each tier is secured in place using hex nuts along the rods. The assembly is then enclosed in the recovery coupler and closed off on both ends with bulkplates, as shown in Figure 2.4.3. The coupler includes two camera holes for video footage and a hole for the screw switches, so that they can be switched on at the launch pad. The eye nuts on either end provide attachment points for the shock cords that connect to the parachutes. Additionally, the bulkplates include screw terminals that connect the altimeters to the e-matches located in the charge wells, as shown in Figure 2.4.4.

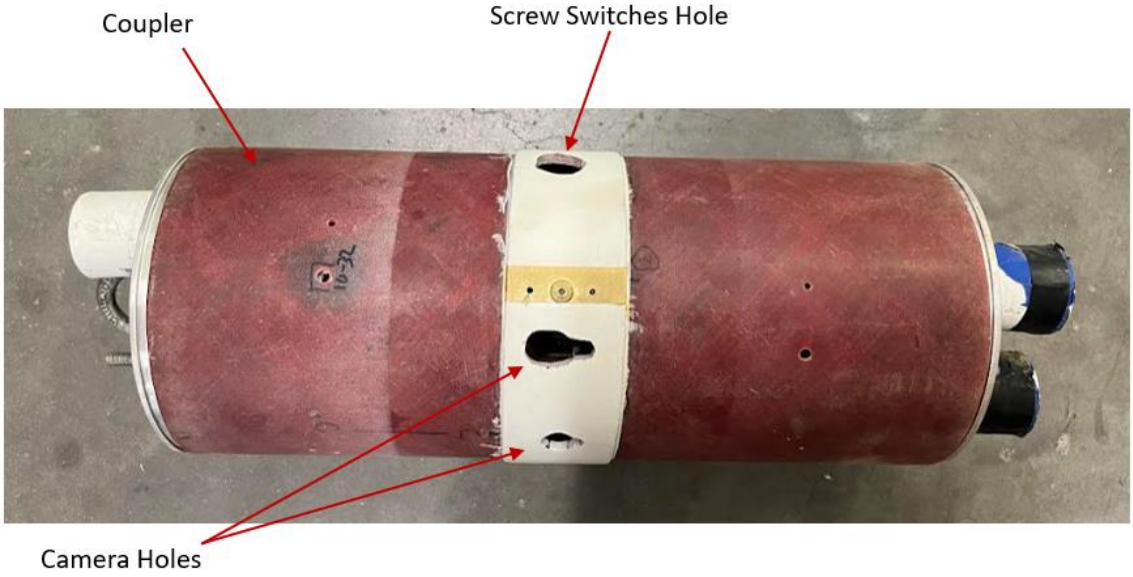


Figure 2.4.3: Recovery Coupler

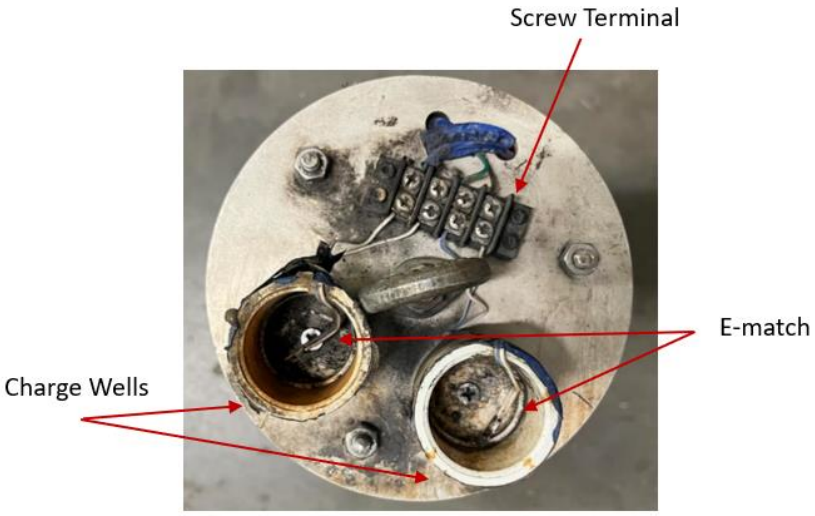


Figure 2.4.4: Recovery Bulkplate

Figure 2.4.5 shows how all components of the system connect. Once the screws are tightened, the batteries power the altimeters. After launching, when the rocket reaches apogee and then the pre-determined main-parachute altitude, the altimeters send a signal to fire. These signals travel through the screw terminals to the e-matches, which ignite the black powder inside the charge wells, creating a controlled explosion that separates sections of the rocket and deploys the drogue at apogee and the main parachutes at a lower altitude to avoid drift.

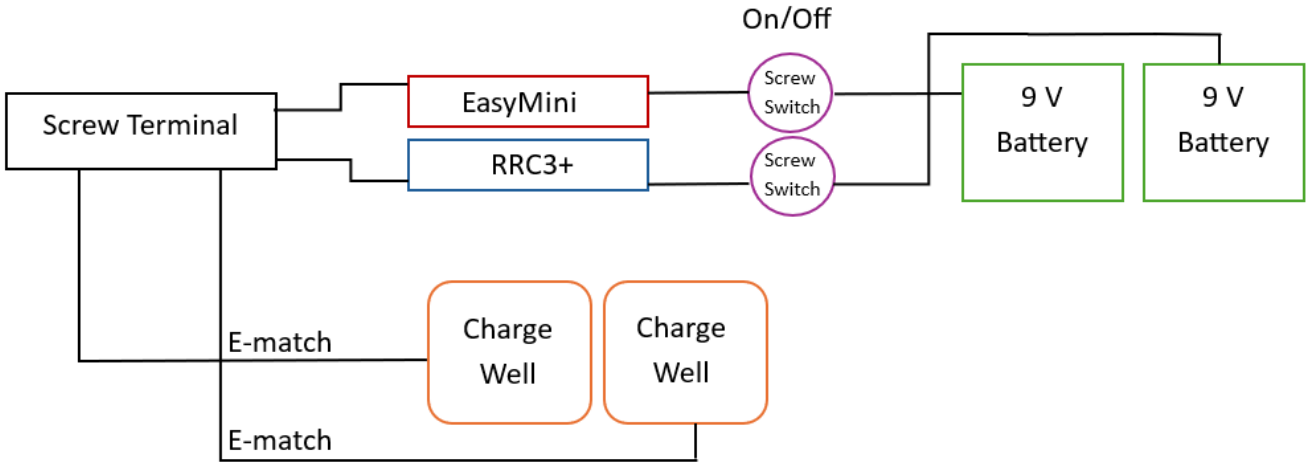


Figure 2.4.5: Recovery System Connection Diagram

The Featherweight GPS tracker operates at a frequency of 924.600 MHz and provides real-time tracking of the rocket. The frequency of this GPS falls within the 33 cm technician-class ham

radio band, and while the device's documentation suggests that it has been pre-licensed for public use via CFR 47 part 15, the team acquired ham radio licenses for safety. Figure 2.4.6 shows the rocket's position in three dimensions (x,y,z) during flight, obtained from the integrated tracking system.

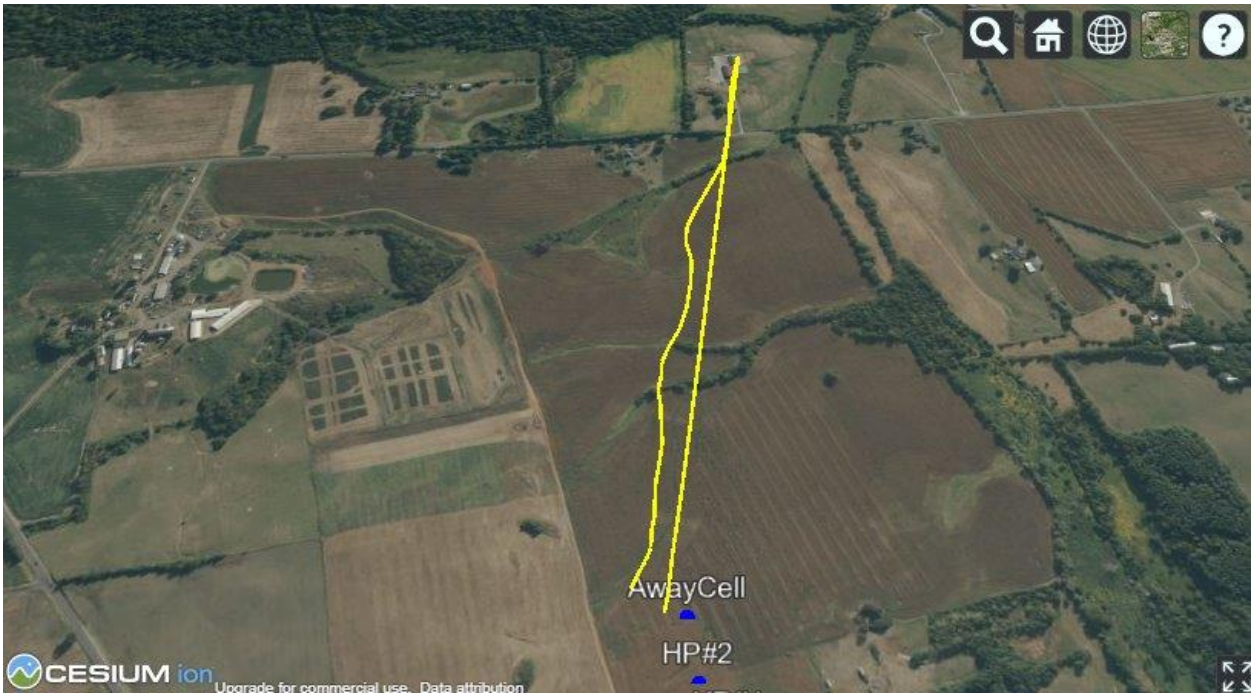


Figure 2.4.6: Featherweight GPS Tracker Results

Figure 2.4.7 shows the Featherweight GPS tracker and its ground station interface, which also shows live flight data. This data was used to cross-check measurements from the airbrake avionics sensor suite discussed in section 2.6, which helped to validate the recorded values.

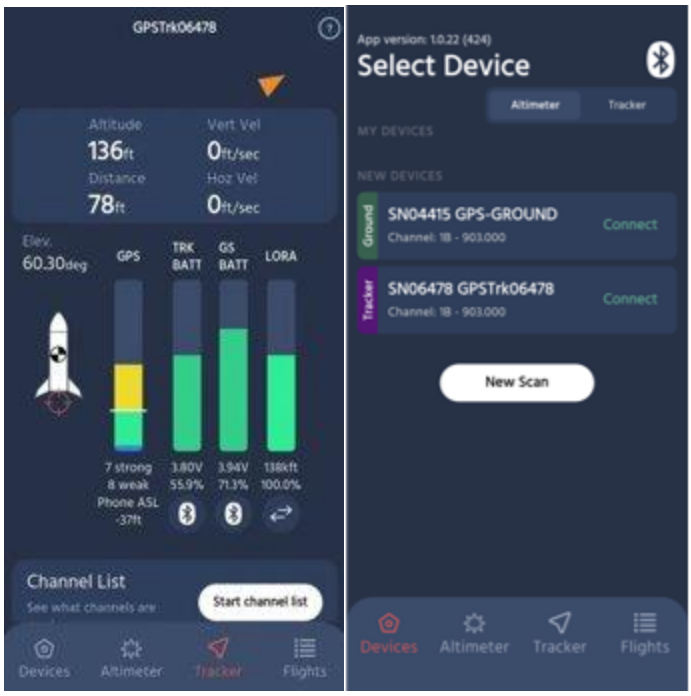


Figure 2.4.7: Featherweight GPS Tracker and Ground Station Interface

Table 2.4.1 Recovery: Status of Components

Item	Component name or description	To be fabricated or purchased	Detailed design completed or purchase specified	Design fabricated or purchase received
1	GPS Tracker	To be purchased	Purchase specified	Received
2	Primary Altimeter	To be purchased	Purchase specified	Received
3	Secondary Altimeter	To be purchased	Purchase specified	Altimeter ordered
4	Batteries	To be purchased	Purchase specified	Received
5	Electrical Wire	To be purchased	Purchase specified	Received
6	Battery connectors	To be purchased	Purchase specified	Received
7	Battery and Camera Tiers	To be fabricated	Detailed design complete	Fabricated
8	Avionics Tier	To be fabricated	Updated detailed design in-progress	Pending redesign completion
9	Charge wells with hardware	To be purchased	Purchase specified	Purchase received
10	Black powder	To be purchased	Purchase specified	Purchase received

11	Screw terminal blocks	To be purchased	Purchase specified	Purchase received
12	E-matches	To be purchased	Purchase specified	Purchase received
13	Coupler (sizing, leveling, holes)	To be fabricated	Detailed design complete	Fabricated
14	Shock cords	To be purchased	Purchase specified	Purchase received
15	Parachutes	To be purchased	Purchase specified	Purchase received
16	Swivel links, eye nuts	To be purchased	Purchase specified	Purchase received

2.5 Airbrake Mechanism & Supporting Structures

The airbrake mechanism is housed inside the aftmost coupler, between the fin can and lower body tube.

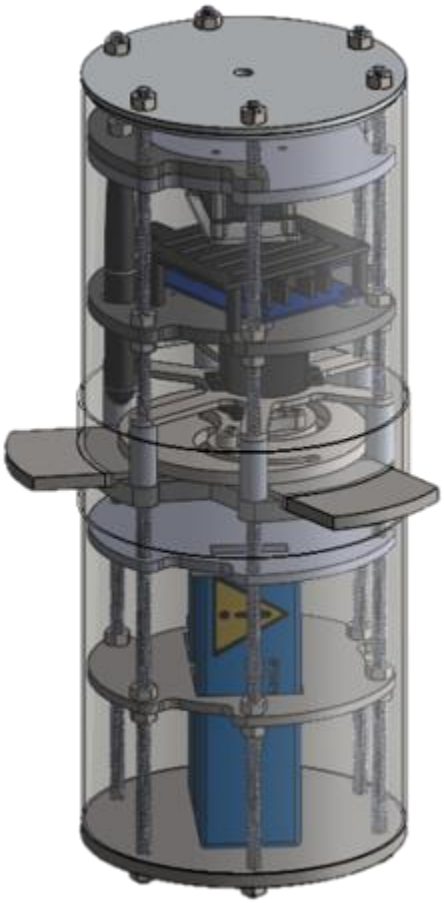


Figure 2.4.1: Airbrake Coupler Assembly

In Figure 2.5.1, the coupler is shown as transparent. The bulkplates on either end hold the internal assembly in place inside the rocket. The brake mechanism is suspended midway between the upper and lower bulkplates, with electronics above and batteries below. The electronics and batteries are held on tiers suspended in the same way as the airbrake mechanism, which were 3D-printed to reduce cost.

To maximize the diameter of the spiral plate and thereby the displacement of the brakes, six small threaded rods around the edges of the spiral plate were used to fixture the internals. ¼”-20 rods were selected to provide the necessary strength with minimal diameter. This is illustrated with a top-down view of the coupler in Figure 2.5.2.

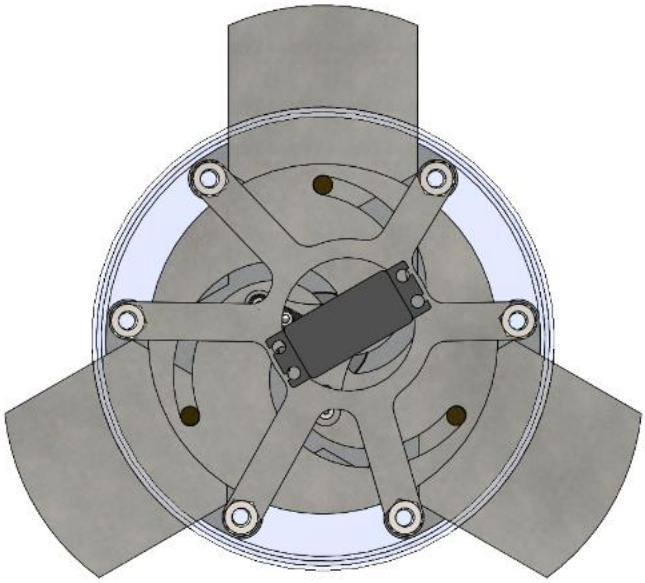


Figure 2.5.2: Top-Down View of Airbrake Mechanism

Nyloc nuts were chosen to hold the airbrake assembly and the bulkplates in place due to their resistance to vibration, and other component tiers were fixtured with regular hex nuts to make assembly faster and easier.

The motor was mounted above the spiral plate due to the need for a coupling assembly to convert the motor’s spline shaft to a ½” hex shaft for better shear strength through the spiral plate, which would otherwise collide with the brake plates and guide plate. This also allowed for a flange bearing to be embedded in the guide plate opposite the motor to ensure axial stabilization. However, it meant that an additional bracket would need to be suspended above the spiral plate to hold the motor. The motor’s spline profile was not replicable with the tools available, so the included arm attachment was used to connect the motor to the mechanism. A converting piece was designed to attach the motor arm to the spiral plate to keep the latter level, because the

motor arm offset the attachment point from the center of rotation. This converting piece also held the upper end of the hex shaft in place

The final airbrake mechanism concept is shown in Figures 2.5.3 and 2.5.4.

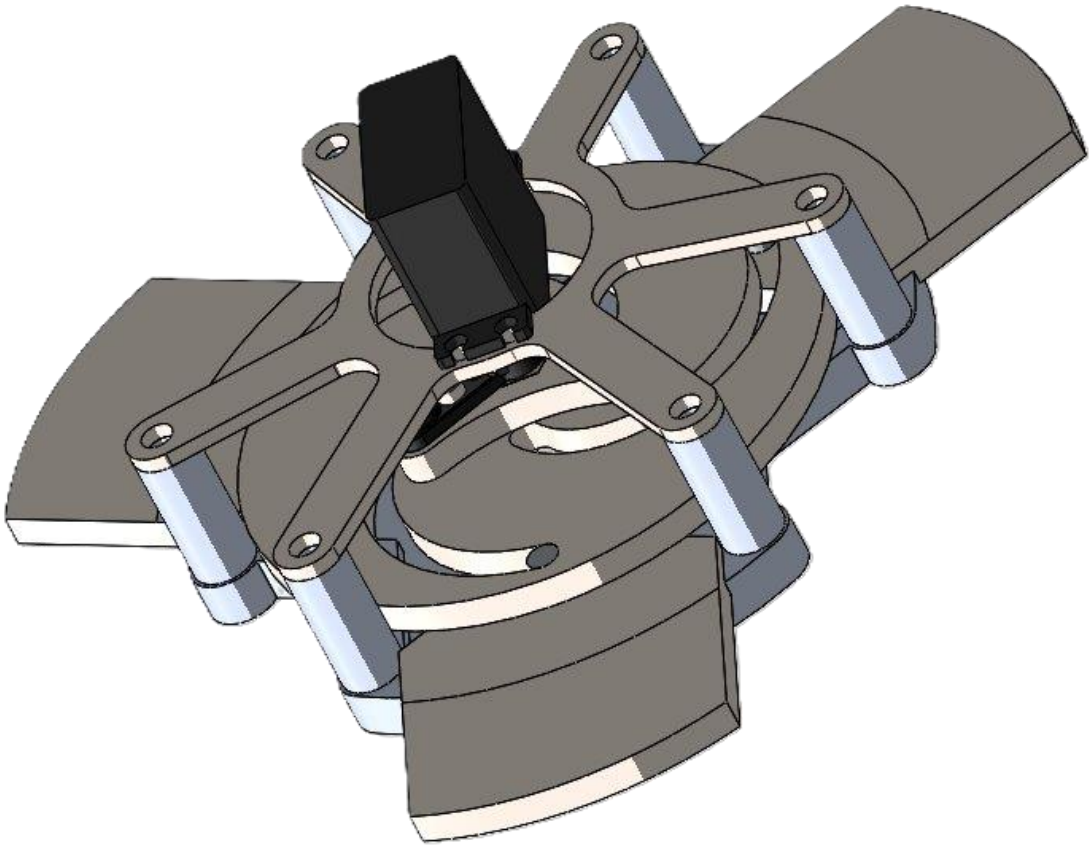


Figure 2.5.3: Airbrake Mechanism Assembly CAD

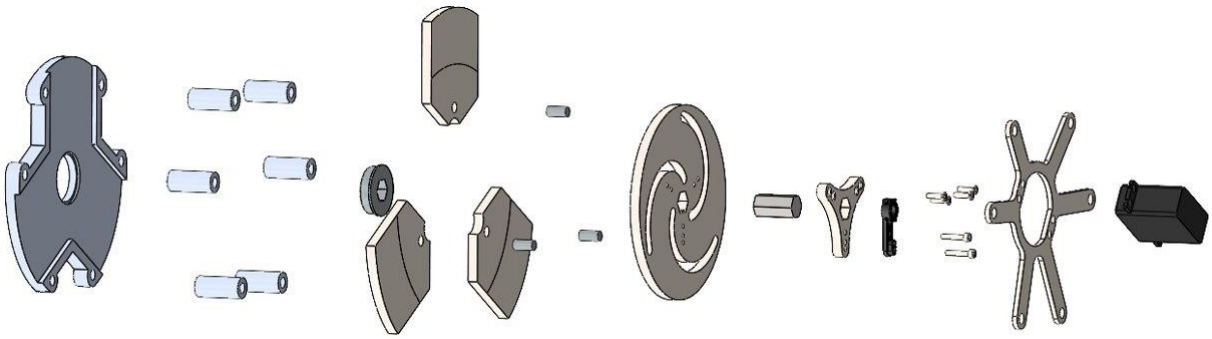


Figure 2.5.4: Exploded View

Manufacturing drawings for all of the parts that were fabricated for the project, including brake components and coupler bulkplates, are included in Appendix 9.1.

To reduce friction and material wear, either the brake plates or the guide plate were to be made of aluminum, and the other steel. It was decided that the brake plates should be steel, for two reasons. Firstly, their geometry lends itself to being laser-cut very easily, creating a two-step start-to-finish manufacturing process including surface finishing. The guide plate would involve a more complicated process regardless. Secondly, the strength of steel helped to ensure that the inner edge of the brake plates would not fail under the shear load imposed by the pin connection.

Other material choices were made primarily due to manufacturability, but some parts which see higher stresses were specified to be made of steel for strength, namely the motor crank arm transmitter which sits between the hex shaft and the motor crank arm in the exploded view, and the pins. The former was laser-cut out of mild steel, and the latter was made of stainless steel. Due to the 2-dimensional nature of the spiral plate and the motor attachment bracket shown toward the right in the exploded view, these were also laser-cut from mild steel.

Reducing the weight of the mechanism was not a concern. The rocket design called for roughly 15 pounds in the airbrake coupler for stability, and only a portion of this weight could be moved around or removed before negatively affecting the rocket. It would be inadvisable to put all of the necessary weight (up to 15lbs) in the nose cone coupler for fear of stressing the fiberglass aerostructure under bending. Therefore, the airbrake coupler weight should remain relatively high, and the exact CG could be controlled with smaller weights in the nose cone if necessary. Preliminary estimates in SolidWorks placed the coupler weight at approximately 10 pounds including major electronics such as the LiPo batteries and motor. Once assembled in its entirety, the actual system weighed 12.083 pounds. The airbrake mechanism as it appeared shortly after manufacturing is shown in Figure 2.5.5.



Figure 2.5.5: Airbrake Mechanism Assembly

Figure 2.5.5 also shows the configuration used for the initial motor calibration, to ensure that the software knew how far the brakes were extruded for any motor angle.

Figures 2.5.6 a/b show the full airbrake coupler internal assembly as it appeared during the full-scale launch tests, discussed in sections 3.8 and 3.9.

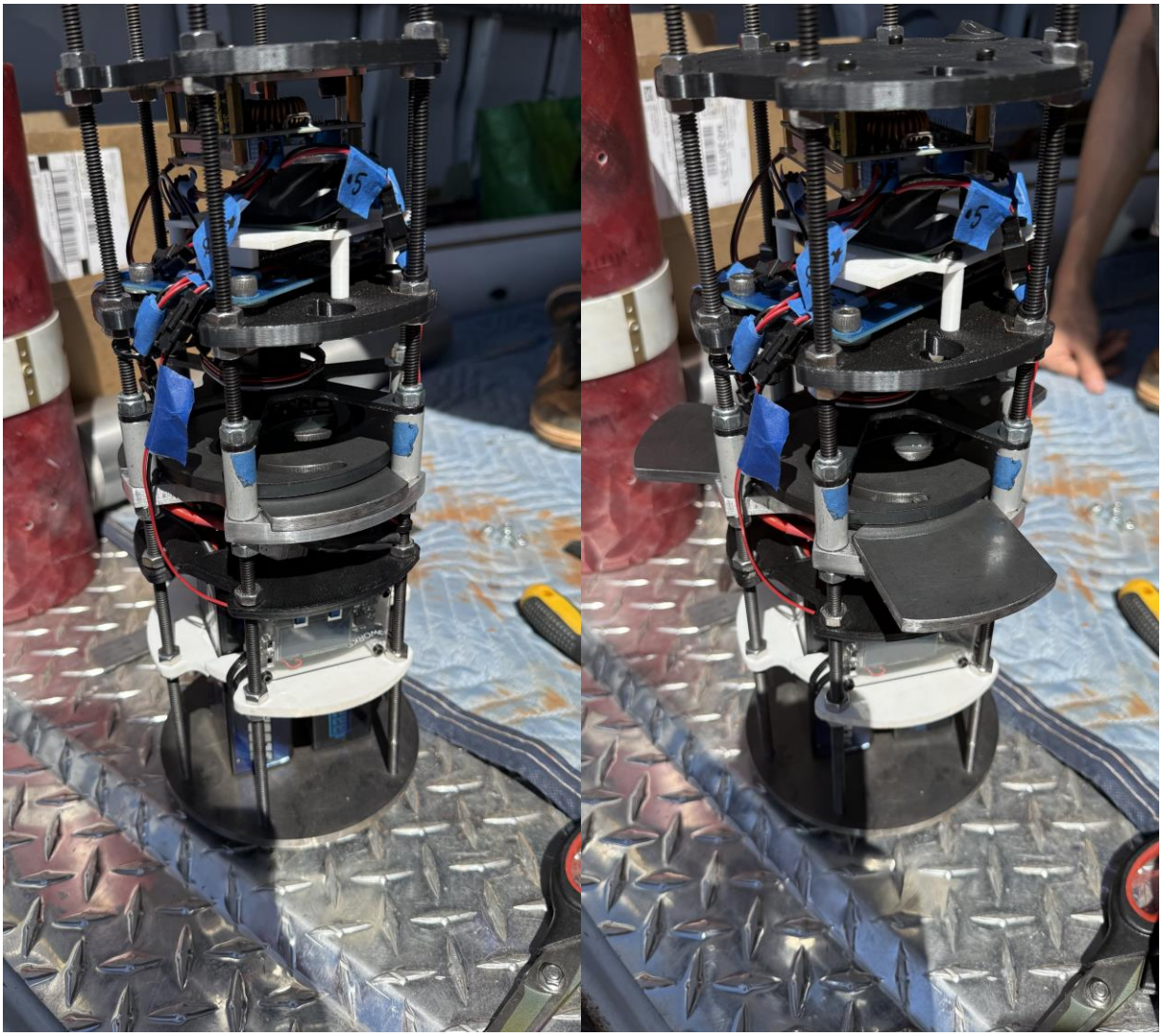


Figure 2.5.6 a: Brakes Retracted b: Brakes Fully Deployed

To ensure that the mechanism would not bind as it deployed and retracted, WD-40 was applied to each sliding zone. WD-40 was chosen due to its low viscosity, avoiding viscous drag but still allowing the brakes to slightly hydroplane on the guide plate surface.

Table 2.5.1 Airbrakes: Status of Components

Item	Component name or description	To be fabricated or purchased	Detailed design completed or purchase specified	Design fabricated or purchase received
1	Airbrake mechanism	To be fabricated	Detailed design complete	Fabricated, assembled

2	Retention system (rods, bulkplates)	To be fabricated	Detailed design complete	Fabricated, assembled
3	Battery retention plates	To be fabricated	Detailed design complete	Fabricated
4	Coupler (sizing, leveling, slots)	To be fabricated	Detailed design complete	Fabricated
5	Lubricant	To be purchased	Purchase specified	Purchase received
6	Motor	To be purchased	Purchase specified	Purchase received
7	Retention hardware	To be purchased	Purchase specified	Purchase received
8	Mechanism assembly hardware	To be purchased	Purchase specified	Purchase received

2.6 Airbrake Avionics

The airbrake avionics subsystem was designed to collect flight data, estimate the rocket's state space, and command airbrake actuation during flight. This subsystem includes the physical flight computer electronics and the software used to control it. The electronics are housed inside the airbrake coupler in multiple tiers, with the main processor being mounted above the airbrake mechanism, and the batteries being mounted below.

The subsystem is divided into four main functions: sensing, computation, actuation, and power distribution. The sensors measure the rocket's acceleration, gyroscopic motion, and altitude. The microcontroller processes these measurements and estimates the state space. The actuator receives commands from the controller and physically deploys the airbrakes. The power system provides regulated voltage to both the low-voltage flight computer and the high-voltage servo through two different regulators, mounted above and below the brakes.

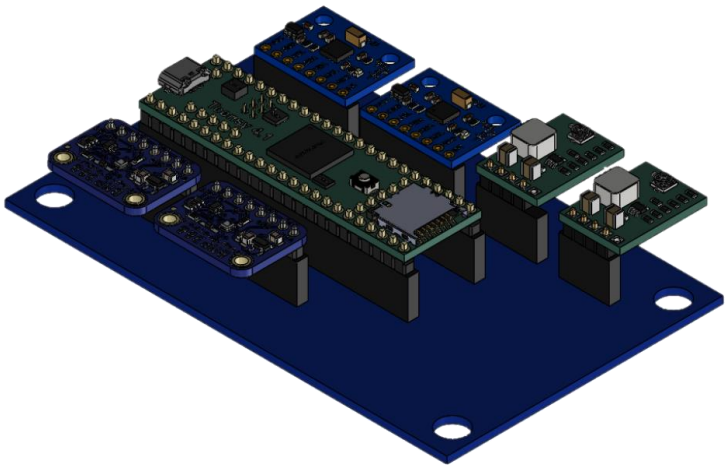


Figure 2.6.1: Custom PCB Design

Figure 2.6.1 shows the custom-printed circuit board used for the airbrake flight computer. The PCB was designed to reduce loose wiring, reduce vibration-related risks, and provide fixed electrical connections between the Teensy 4.1, sensors, power inputs, and servo signal output. This was important because loose jumper wiring could create false sensor readings, intermittent power loss, short circuits, or full system failure during launch.

The flight computer is built around a Teensy 4.1 microcontroller. The Teensy 4.1 was selected because the airbrake controller requires fast sensor reading, real-time filtering, state estimation, data logging, and servo command output. The benefit of the Teensy 4.1 is the capability to do high-speed processing specifically for complex computation, while being easy to prototype. The microcontroller receives data from two barometric pressure sensors (BMP) and two inertial measurement units (IMU). The use of two sensors for each measurement type provides redundancy and allows the software to fuse sensor readings instead of relying on a single sensor for greater accuracy.

The BMP388 barometers measure atmospheric pressure, which is converted into altitude. Altitude is one of the most important state variables because the primary function of the airbrake controller is to predict whether the rocket will exceed the target apogee. The IMU measures acceleration and gyroscopic motion. These measurements are used to detect flight stage, estimate state space, coast conditions, and determine whether airbrake deployment is safe.

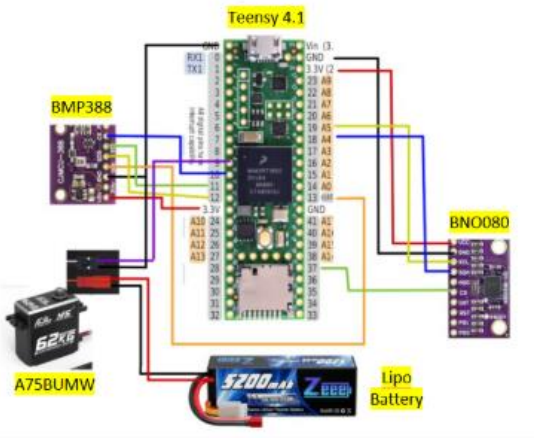


Figure 2.6.2: Wiring Diagram

The actuator for the subsystem is the A75BUMW high-torque servo. This servo was selected due to its high-performance capabilities and ability to handle the expected flight conditions. The servo converts the command from the flight computer into rotational motion, which drives the airbrake mechanism. Because deployment occurs during the relatively short and high-velocity coast phase of flight, the servo must be able to move the brake mechanism quickly while resisting aerodynamic loading on the airbrake plates.

Power is supplied by two 3S LiPo batteries, which provide energy to the flight computer and servo system. Since the Teensy and sensors cannot operate directly from the full battery voltage of 22.2V, Buck converters are used to step the voltage down to usable levels. A DROK 5V step-down converter supplies the microcontroller and sensor electronics, and a separate DROK 16.8V step-down converter supplies the high-torque servo. Separating the servo power supply from the flight computer power supply reduces the risk that servo current spikes will reset the microcontroller or interrupt sensor readings, which was an issue that was identified during pre-flight testing of the airbrakes.

The avionics subsystem was manufactured and integrated in several stages. First, the PCB layout was designed to match the selected components and wiring requirements. The custom PCB was then purchased and assembled with headers, connectors, and wiring interfaces. Next, the Buck converters were connected and adjusted to provide the required 5V and 16.8V outputs. The sensors, Teensy 4.1, servo signal line, and power rails were then connected according to the wiring diagram. Finally, the complete system was installed into the airbrake coupler and tested with the flight software to verify sensor readings, data logging, servo movement, and controller response.

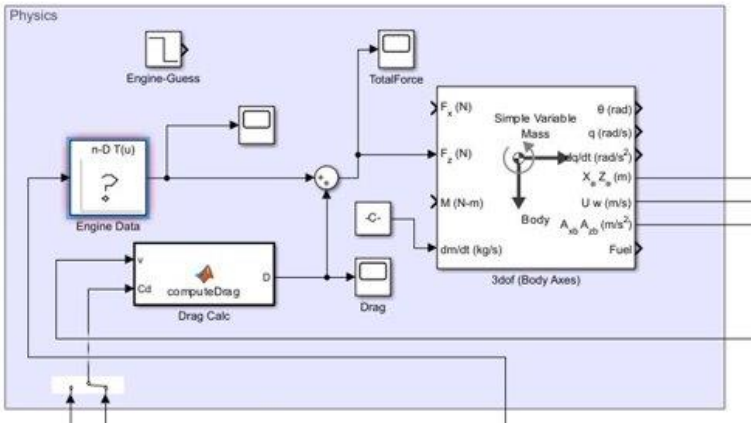


Figure 2.6.3: Controller and Simulink

The flight software determines how the electronic hardware is used during flight. The software architecture is used for flight-state estimation and subsequent airbrake control. At startup, the system initializes the sensors, records the baseline pad altitude, and prepares the data-logging system. During launch, the software detects acceleration above a threshold and begins identifying the rocket's flight state. During coast, the controller fuses sensor data, filters altitude and acceleration measurements, estimates velocity, predicts final apogee, and commands the servo to deploy the airbrakes by the required amount. Finally, once apogee is met, the airbrakes are retracted to ensure safer landing conditions.

The controller uses a feedback loop structure. The estimated current altitude, vertical velocity, acceleration, and drag equations are used to predict the rocket's max apogee. The predicted apogee is compared to the target apogee. If the predicted apogee is above the target, the controller increases airbrake deployment to create more drag. If the predicted apogee is closer to the target, the controller reduces deployment. Safety logic prevents deployment during flight states, such as before launch or during ascent.

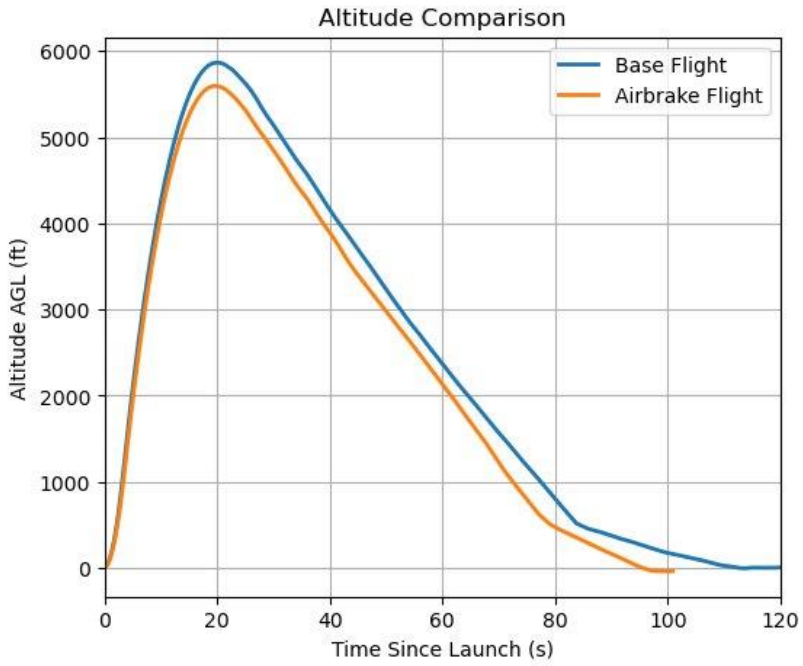


Figure 2.6.4: Flight Test Results

Figure 2.6.3 shows the final flight result graph. The graph demonstrates the relationship between recorded altitudes from the flight computer during the final two launches. The base flight shows that the sensor suite can record and compute the state space of the rocket during flight and proves that the flight computer is operational. Then the airbrake flight shows a decrease in apogee when the airbrakes are deployed, proving that the flight computer successfully commanded the airbrake mid-flight and reduced the apogee of the rocket.

Table 2.6.1 Airbrake Avionics: Status of Components

Item	Component name or description	To be fabricated or purchased	Detailed design completed or purchase specified	Design fabricated or purchase received
1	Teensy 4.1	To be purchased	Purchase specified	Received
2	BMP388 x 2	To be purchased	Purchase specified	Received
3	BNO080 x 2	To be purchased	Purchase specified	Received
4	A75BUMW High Torque Servo	To be purchased	Purchase specified	Received
5	Drok 5V Step Down Converter	To be purchased	Purchase specified	Received

6	16.8 V Step Down Converter	To be purchased	Purchase specified	Received
7	LiPo Battery x 2	To be purchased	Purchase specified	Received
8	Custom Printed Circuit Board	To be purchased	Purchase specified	Received

2.7 Rocket Launch & Recovery Operations

Several days prior to launch, the propellant is mixed and cured, and the motor case is pre-assembled offsite and transported by a licensed individual. If multiple launches are to be done in one day, the propellant reload is placed in a separate case for safety.

Before transporting the rocket to the launch site, the parachutes are folded and packed and the shock cords are bundled, to prevent tangling during deployment.

Prior to launch, the charge wells are filled with black powder and packed, and e-matches are installed and wired to the altimeters. The GPS and airbrakes are turned on, and the couplers are closed and fastened. The shock cords and parachute deployment bags are attached to the coupler bulkplates. The airframe is assembled, and hard and shear points are installed. The rocket is transported to the launch rail, altimeters and cameras are turned on, and an igniter rod is installed in the engine and wired to the launch site's firing box.

After launch, the rocket is located via GPS and the altimeters, cameras, and airbrakes are turned off. The shock cords are disconnected, and they and the parachutes are packed inside the body tubes. Then, the airframe components are carried back to the launch site. If another launch is scheduled, the motor is rebuilt with the propellant reload and all steps are repeated. Otherwise, the launch process is complete.

A flowchart of launch day operations is included in Appendix 9.2.

3. Analysis, Testing, and Evaluation

3.1 Overview

Testing of each subsystem prior to launch days included physical design testing, simulations such as CFD and controller simulations, and flight test analysis using flight data collected from the flight computer. Computational analyses were done to ensure that the design of the rocket and airbrakes could accomplish the goal. This included FEA, CFD, and MATLAB Simulink simulations, which evaluated the structural strength, aerodynamic behavior, and airbrake control performance before flight. These tests demonstrated that the design could handle flight conditions and reduce the max apogee through airbrake deployment. Experimental testing, including static fire testing, switch testing, and launch testing, was used to confirm that the physical system operated properly under realistic conditions. These tests verified that the completed design satisfied the functional requirements related to structural integrity, propulsion performance, recovery reliability, airbrake control, and apogee reduction. The testing identified areas of failure and design risks but also validated the rocket design. Table 3.1.1 summarizes the final status of each major analysis and test, including the functional requirements the tests aim to accomplish and the methodology used.

Table 3.1.1 Final Status of Testing and Evaluation

	Analysis or Testing	**FR(s)**	**Test methodology**	**Status or Outcome**
1	Brake Plate Finite Element Analysis	Must withstand 22 lbf drag across brake surfaces	SolidWorks FEA was used. Capable of handling 1600 lbf of force, which is far above the minimum of 22 lbf.	Completed. The FEA proved that the airbrake design can handle real flight conditions as well as a CATO/crash.
2	Propulsion Static Fire Testing	Student researched-&-developed rocket engine	Static fire was done using a pressure transducer and load cell.	Completed. The static fire showed similar results to those predicted by simulation.
3	Toggle Switch Resistance Test	Rocket must land safely and be reusable	High-pressure airflow was directed onto the toggle switches. Airflow did not affect the switches.	Completed. Demonstrated the feasibility of switches and simulated airflow conditions.

4	Control System Simulation	Live-feedback autonomous control system	A MATLAB Simulink model was developed using motor thrust, mass, gravity, body drag, and airbrake drag. Velocity-delta and predictive guidance control methods were simulated and compared.	Completed. Controllers were designed and simulated to prove a reduction in apogee using airbrakes.
5	Rocket and Airbrake Computational Fluid Dynamics	Airbrakes must reduce apogee by some degree	CFD simulations were performed on simplified rocket geometry at 0%, 50%, and 100% airbrake deployment. Drag force and drag coefficient were calculated for each case.	Completed. Three values of drag were computed and used in flight computer software and simulation.
6	Subscale Airframe Flight Test	Volume constraints, Rocket must land safely for reuse	A subscale launch was conducted on February 28, 2026 using an M1186 motor. Successfully launched and recovered the rocket and demonstrated completion of FR's.	Completed. Demonstrated successful launch and tested all electronics. Identified points of failure in the recovery system to be fixed before full-scale launches.
7	Full-Scale Airframe Flight Test	Volume constraints, Rocket should reach a base apogee of at least 4500', Rocket must land safely for reuse	Full-scale launch testing was conducted to evaluate the complete rocket system before or alongside active airbrake validation. Flight data was compared against expected performance.	Completed. Highly successful launch for all systems except airbrakes.
8	Airbrake Flight Test	Airbrakes must reduce apogee by some degree	Three launches were used to evaluate airbrake performance. Final effectiveness was determined by comparing	Completed. Demonstrated an apogee reduction of 4.6%.

			a full-scale baseline launch to a full-scale launch with active airbrakes. Demonstrated a reduction in apogee.	

3.2 Brake Plate Finite Element Analysis

To predict how much force the brakes would be able to withstand, a series of SolidWorks FEA simulations was run. The objective was to perform a simulated test to failure (minimum FOS of 1), to determine the load at which the brakes may begin to fail, and where they would do so.

The test that achieved this involved a 1600 lbf loading across the exposed brake surfaces and gravity. The internal surfaces of the six spacers, guide plate holes, and cover bracket holes were fixed, as the actual mechanism would be fixed inside the coupler. The mesh contained 10.4 million tetrahedral elements and a growth rate of 1.4. A representative segment of the mesh is shown in Figure 3.2.1, and the simulation result is shown in Figure 3.2.2.

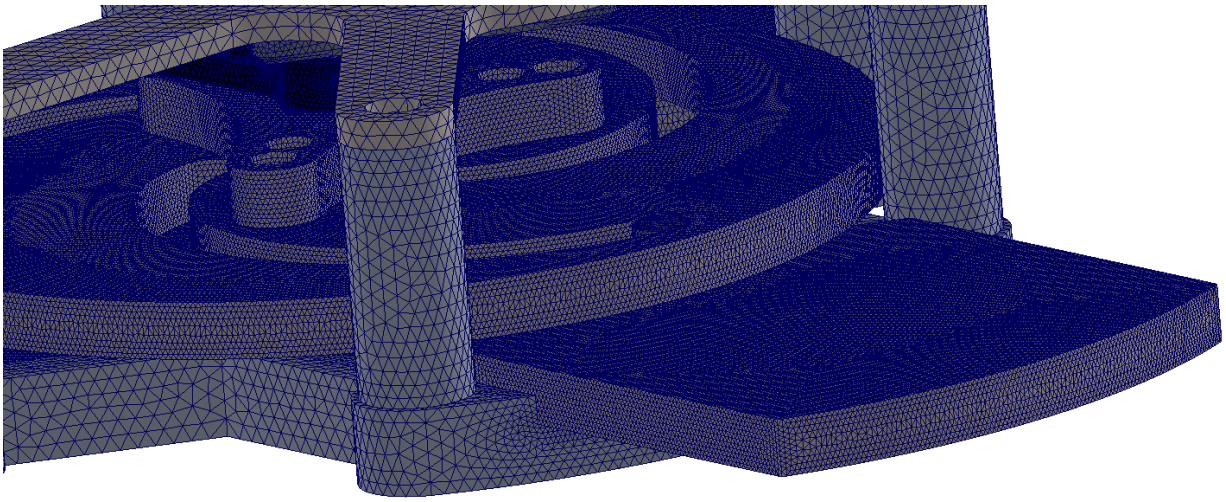


Figure 3.2.1: FEA mesh

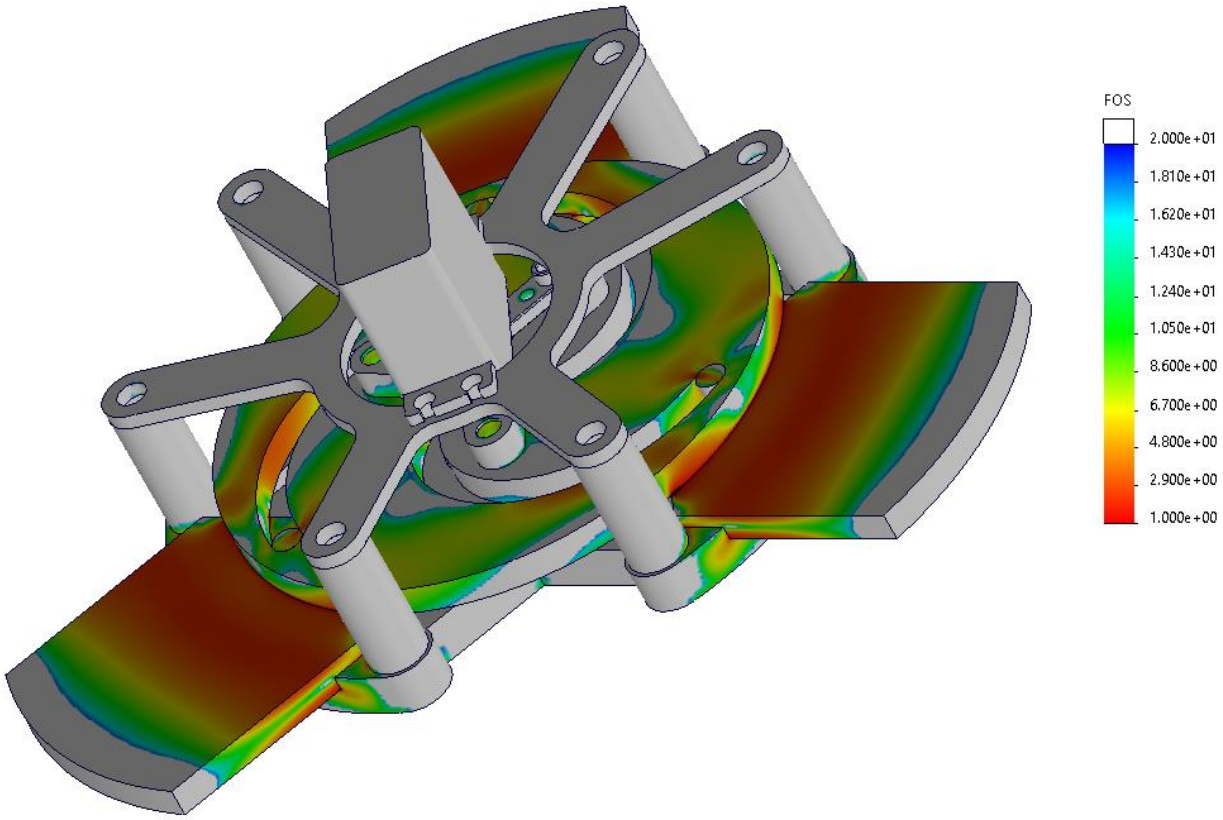


Figure 3.2.2: Test to Failure Factor of Safety Plot

Discarding low-resolution edge effects, this produced a minimum FOS of 1. In Figure 3.2.2, regions with a FOS greater than 20 are displayed gray. The region of likely failure, at the horizontal edges between the brake plates and spiral plate, is shown in Figure 3.2.3.

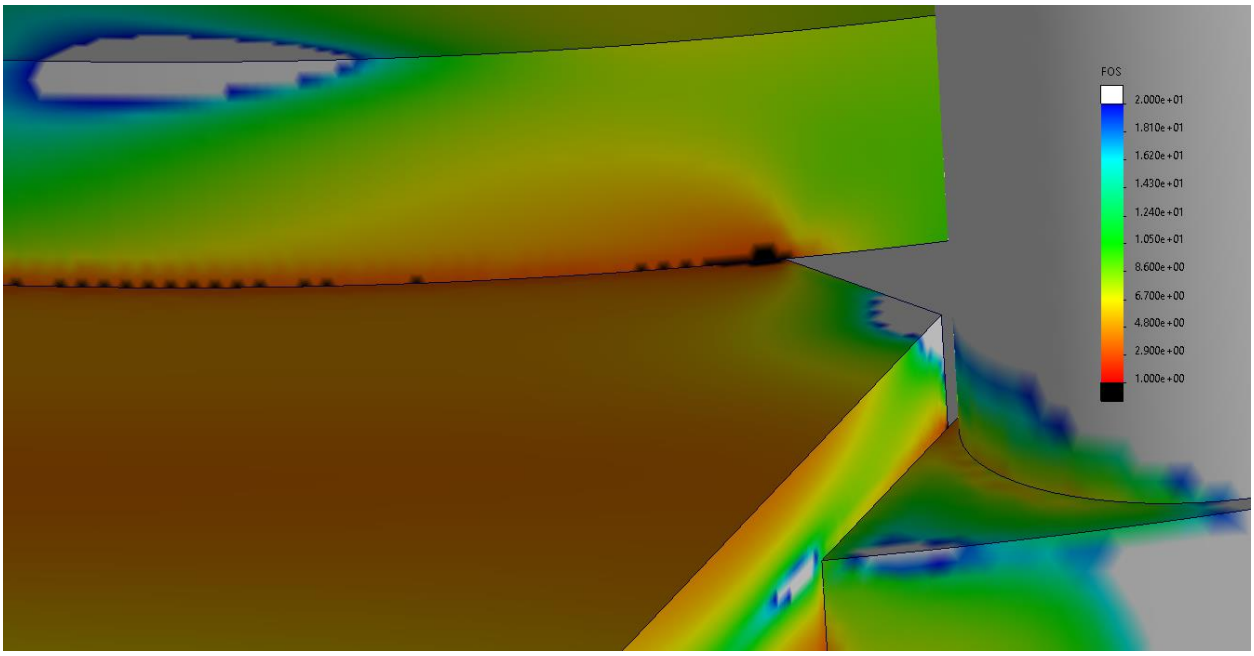


Figure 3.2.3: Failure Region

Edge artifacts measuring one to two elements wide are visible in Figure 3.2.3, which are not considered significant. However, a larger pattern is visible that shows the stress concentration around the edge of the brake plate, on the surfaces of both components. This pattern is visible on all six edges.

Based on this result, these brake plate edges as well as the lower edge of the spiral plate were filed and sanded to a smooth fillet to reduce stress concentration and prevent binding, which created a distinguishable reduction in the amount of force needed to actuate the brakes, even under no load.

While the primary motivation for making the brakes so strong was increased weight rather than the need for strength, the high force predicted to cause yielding indicated that the brakes would be safe even in a malfunction such as a crash-landing, failure to retract at apogee, or catastrophe at takeoff (CATO). This was highly desirable due to the brakes' long procurement and manufacture time, and consequently lessened the risk of flying the rocket's first subscale launch with the brakes onboard. The decision to go forward with that plan pushed the timeline forward and allowed for multiple later attempts at active braking during flight, by simultaneously testing the brakes' data reading and processing with the recovery system and airframe (the subscale launch is discussed in detail in section 3.7).

3.3 Propulsion Static Fire Testing

To validate the simulation motor performance as described in section 2.3, experimental testing must be conducted. Static tests are a method utilized to validate an engine's performance without

launching it in the rocket. The static fire tests were conducted at the Tripoli Central Virginia BattlePark Rocket Launch site located at Crittenden Ln, Rapidan, VA 22733. To contain the engine to the ground and minimize any safety threat, a test stand was used. This test stand is designed in modular sections, consisting of a steel base plate and four aluminum X-beams at each corner. A second steel plate, with a cutout matching the motor diameter (98mm) sits atop the beams. This assembly is considered 1 segment, each measuring 1 ft in height. This test stand is designed to be modular and can be extended by an additional 3.5 segments. The assembly is also bottom-heavy and anchored to the ground with steel cables to ensure it stands upright during tests. All members return to the control area to maintain a safe distance from the test stand and minimize potential risk during static fires.

The vertical test stand interfaces with an RMTS (Rocket Motor Test Stand) system, which records both pressure and thrust data. It is activated via a wireless communication system that enables remote activation and data collection. The data collected from the test stand is both pressure and thrust data, ensuring that the engine produces the thrust needed for liftoff and validates that the engine will not overpressure. To ensure that accidental activation of the engine does not occur, the stand has separate arming switches for both the board and the ignition switch (powered by separate batteries). Another factor to consider when running engine tests is atmospheric temperature at the site, since engines may perform differently at varying temperatures. Therefore, the electronics stand also includes a weather monitoring station with a wireless relay to a readout display.

The pressure is collected from the forward closure through an NPT pipe that is connected to a pressure transducer. Thrust is collected via a button load cell at the base of the test stand. Both of these sensors are connected to the RMTS system for remote data transmission.

As shown in Figure 3.3.1, the motor was installed into the test stand and fired. The motor was encased in a ¼” aluminum motor case. On the bottom of the motor case, the graphite nozzle was installed, with an aluminum washer below it to help distribute the thrust back to the case. Below the nozzle washer was a snap ring, which ensured that the nozzle could not dislodge from the case. On the forward side of the motor case (facing the top of the rocket), the propellant was slid into the case, encapsulated in a phenolic liner to help insulate the case from the heat of combustion. On top of the propellant was the forward closure, sealing the engine except for the nozzle. On top of the forward closure was another snap ring to hold the assembly together. Both the nozzle and forward closure had 2 O-rings to help seal any gaps with the motor casing. If the motor were to over-pressurize, the snap rings are designed to give in, allowing for a controlled upward or downward release rather than a complete explosion and disintegration of the case, which would result in shrapnel being expelled in all directions.



Figure 3.3.1: OpenMotor Bates Grain Simulation Results

The results of this static fire were analyzed and compared against the thrust produced within the simulation. As can be seen in Figure 3.3.2, the peak thrust was 1844.1 N with an average thrust of 1627.5 N, which is very similar to the simulation results mentioned in section 2.3. In addition, the total burn time was 3.941 seconds, which is very much in line with the predicted 3.83 seconds of burn time.

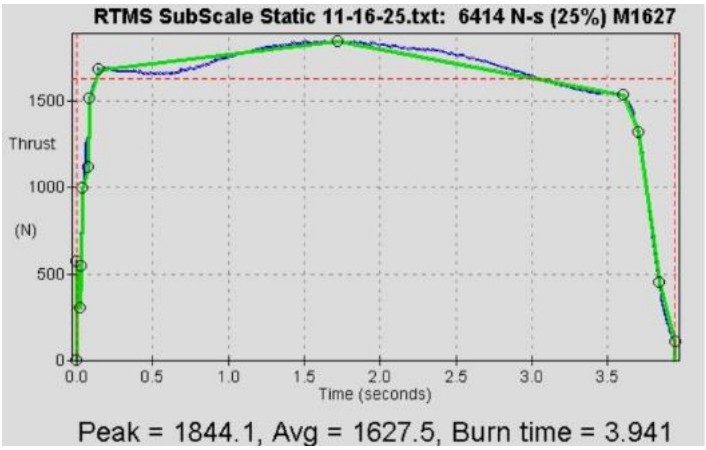


Figure 3.3.2: Experimental Thrust Data

These experimental results gave the team confidence in the simulated flight, including apogee and flight time of the rocket.

3.4 Toggle Switch Resistance Test

The sensitivity of the altimeters, and the risk of a premature separation charge detonation, necessitates that the altimeters be turned on only after the rocket has been lifted to its vertical position on the launch rail. This means that an externally accessible switch must be incorporated. The initial choice was a toggle switch, much like a residential light switch, for cleanliness and ease of use. Two switches were added, one for each altimeter, and were securely adhered to the switchband with Bondo epoxy. The main concern with these switches was whether the flow of air over the switches during flight would cause them to activate, shutting down the recovery systems during flight. To mitigate this, the switch was mounted with the "ON" setting oriented in the downwards direction, allowing aerodynamic forces to enforce that setting.

Although the orientation of the switch was designed to prevent drag forces from switching it off, testing was done to ensure no accidental toggling of the switch by the airflow. This was done by directing compressed air at roughly 60 psi at a switch in an attempt to toggle it. The switch was able to keep its positioning even under this load, ensuring that accidental toggle due to airflow was not a concern. However, for additional safety, when the rocket was mounted vertically, electrical tape was placed over the toggle to prevent it from switching into the off position.

3.5 Control System Simulation

In addition to physical integration testing and full-scale launch testing, simulation was used to validate the airbrake control loop before flight. This test was necessary because the airbrakes could not be repeatedly tested with real flight conditions without launching the rocket, which is an expensive endeavor. The simulation allowed the team to evaluate whether the controller could predict the apogee, command deployment, and reduce the rockets' final altitude. This test directly addressed the functional requirements that the airbrakes must reduce apogee by a measurable amount and that the final system can achieve apogee reduction.

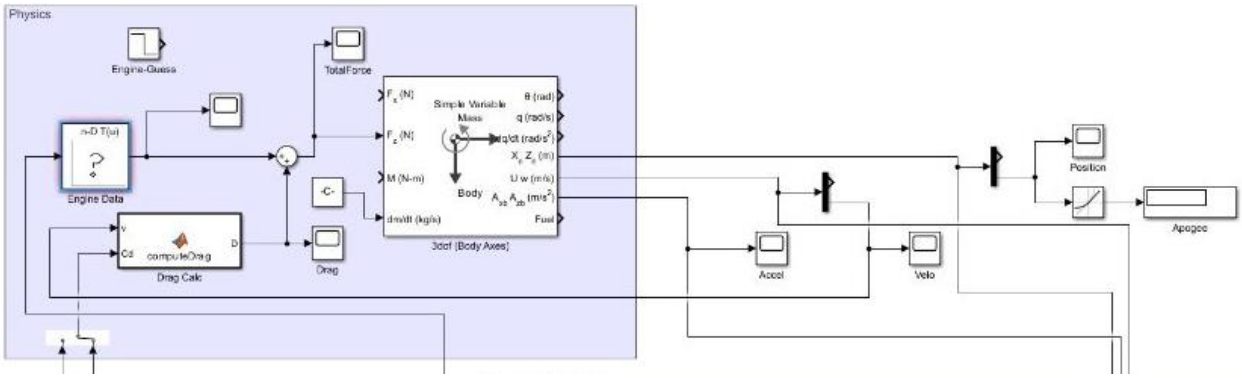


Figure 3.5.1: MATLAB Simulink Flight Simulation Model

The first simulation was done through a custom MATLAB Simulink model to ensure validation of flight physics shown in Figure 3.5.1. The simulation modeled the flight path of the rocket using motor thrust, mass, gravity, aerodynamic drag of the rocket body, and additional drag created from the airbrake deployment. The model was built to replicate the simple flight physics of the rocket without additional airbrake drag, so it can be referenced against flight data and OpenRocket predictions to validate the model. This is important to demonstrate a baseline comparison for the airbrake controller to simulate rocket behavior under expected rocket behavior and determine the effectiveness of the controller and altitude reduction.

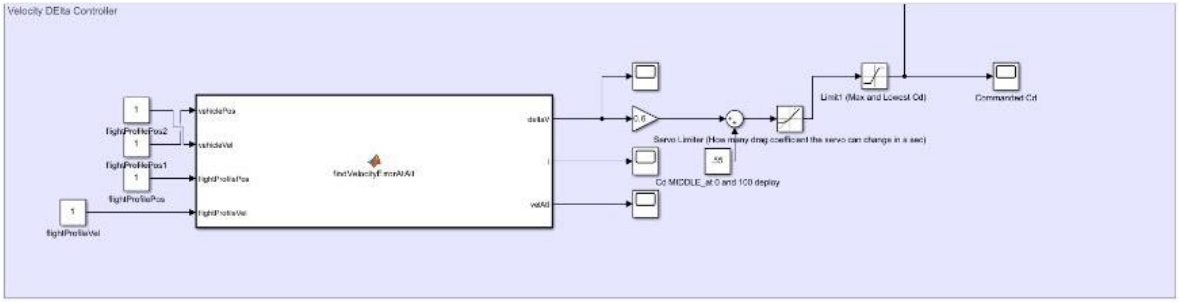


Figure 3.5.2: Velocity Delta Controller

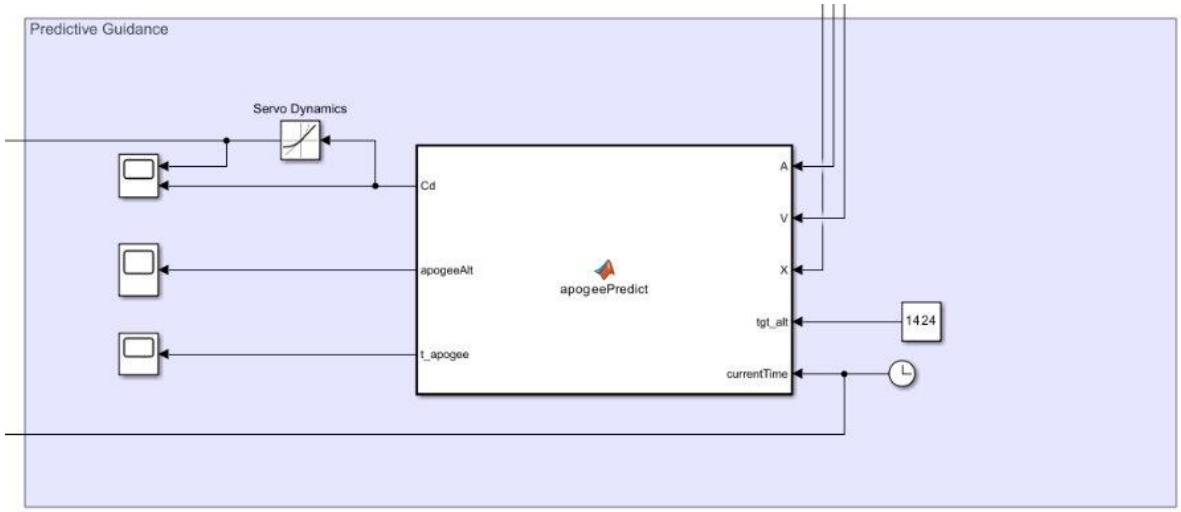


Figure 3.5.3: Predictive Guidance Controller

Once the model was validated, the airbrake controller models were added. A 100% deployment case was simulated to determine the theoretical maximum apogee reduction that the airbrake mechanics could produce to be used as a baseline and reference. Then the control methods were developed in simulation, which are the velocity-delta controller and the predictive guidance controller. Figure 3.5.2 shows the velocity-delta controller, which uses the difference between the current flight and the desired flight path to determine deployment. This is a simple control method that does not have updating controls at timesteps but has one large output. Figure 3.5.3

shows the predictive guidance controller that changes deployment at each time step and constantly keeps predicting the best deployment level but also keeps deploying the most appropriate value without overshooting.

Both controllers were able to reduce the apogee to the expected value with slight differences. However, the predictive guidance controller was the most appropriate for the intended goals of the project because it was dependent on the predicted altitude and velocity of the rocket instead of the velocity, which increased the accuracy and adaptability. This controller method was used as the main reference for the flight computer software.

Overall, the simulation results showed that the airbrake avionics and control design were theoretically capable of reducing apogee and helped shape the software to achieve the same control. The 100% deployment of the airbrake verified the maximum reduction the current airbrake design was capable of during the launch, which could later be used as a reference to validate the accuracy of the model and the CFD results.

3.6 Rocket and Airbrake Computational Fluid Dynamics

The goal of the CFD simulations was to accurately predict the drag on the rocket. The limiting factor in the simulation's accuracy was computational expense, due to the limited processing power available. While professional simulations rely on supercomputing clusters incorporating tens of thousands of cores, the maximum power available for this project was 20 cores. This meant that optimizing this simulation involved significant tradeoffs between mesh density and high-quality simulation parameters versus power and time available.

Several preliminary steps were taken to reduce the computational cost of the simulation. First, the geometry of the rocket was simplified. This involved removing protrusions and coupler and body tube seams, filling voids, and filleting corners. These differences are illustrated in Figure 3.6.1.

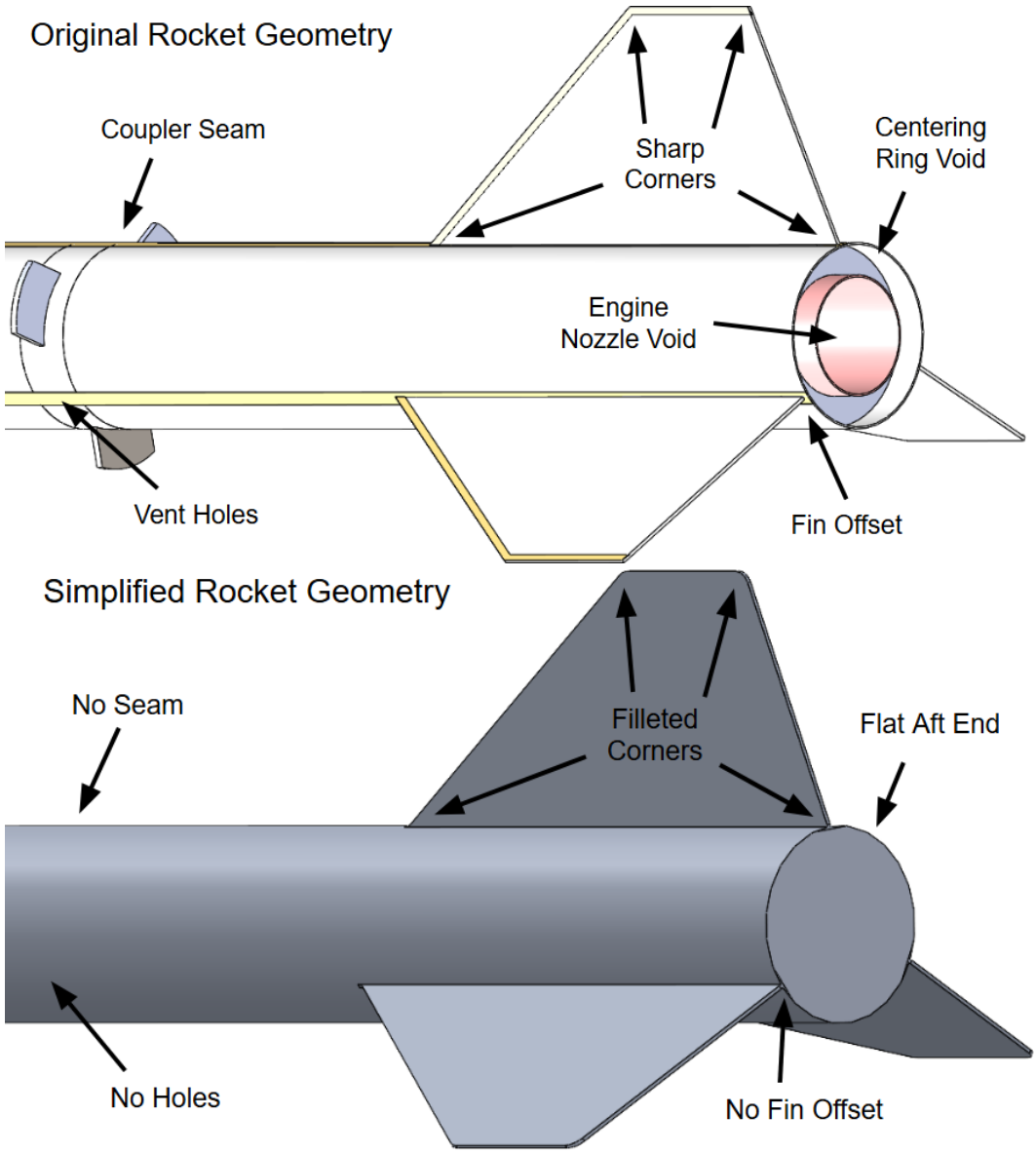


Figure 3.6.1: Fin Can Simplification

The nose cone was also blunted with a spherical tip, as shown in Figure 3.6.2.

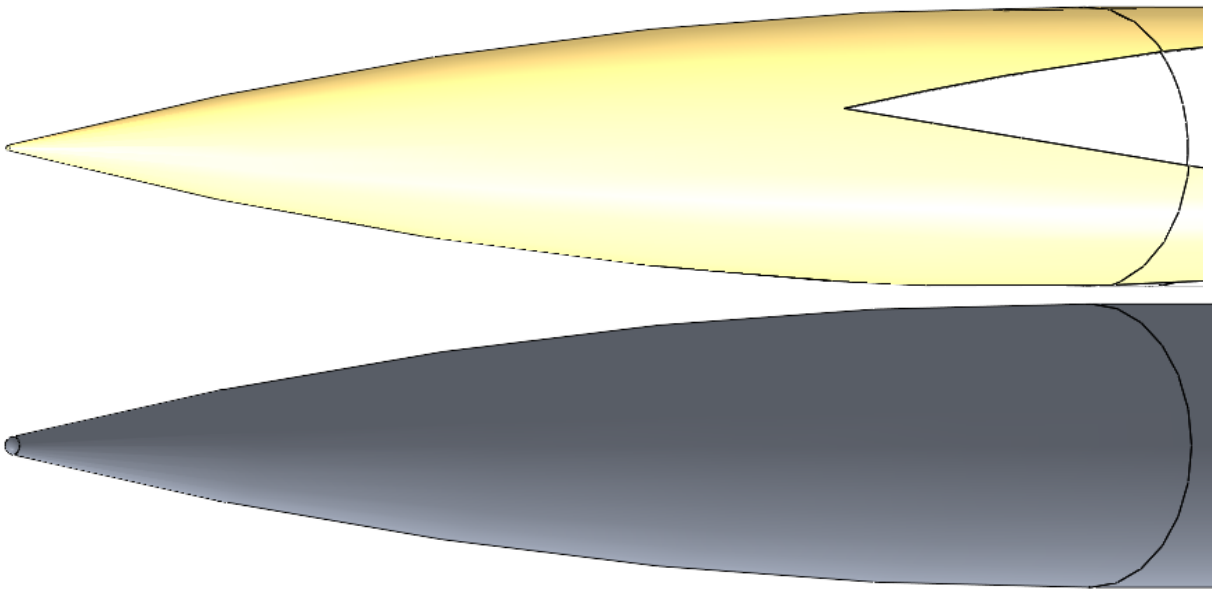


Figure 3.6.2: Nose Cone Simplification

Additionally, the rocket body was split into symmetrical sections of 120°, shown in Figures 3.6.3 and 3.6.4.

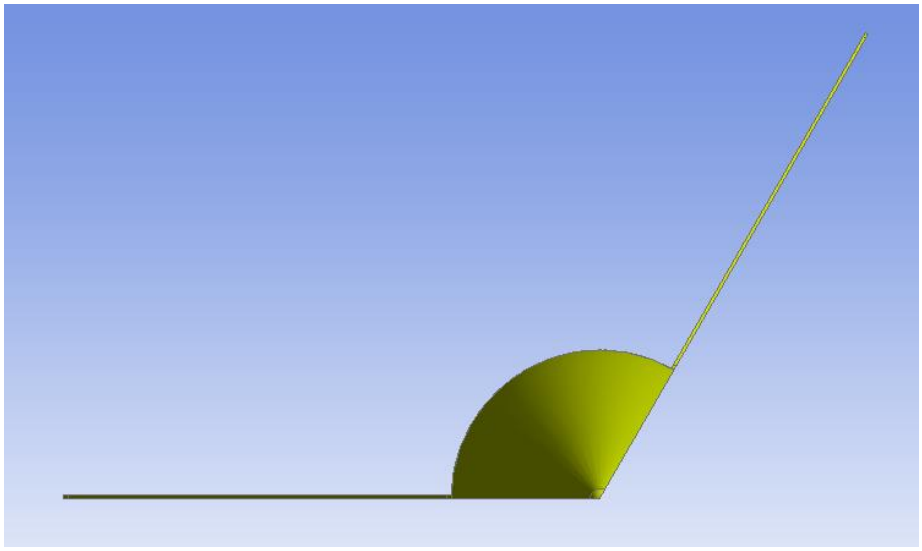


Figure 3.6.3: Rocket Body Section, Front View



Figure 3.6.4: Rocket Body Section, Side View

To maintain the simulation's accuracy, these cuts were made through the fin midplanes rather than where the brakes extrude. The latter is expected to have significantly more complex flow patterns, making the zero-flux symmetry condition less realistic. This configuration allows the resulting control volume to be reduced to one-third at approximately no cost to accuracy.

Enclosure dimensions were chosen based on the rocket body diameter D = 6 inches, and using a simulation done at West Virginia University [1] as a reference. Consultation with Ibrahim Yildiran of GWU's Laboratory for Computational Physics and Fluid Mechanics yielded minimum cushion dimensions of 10 diameters frontal, ±10 diameters lateral, and 30 diameters rear. Due to the high speed of the rocket, excess cushion was used to ensure validity. The dimensions used were 20 D frontal, 20 D radial, and 40 D rear. An inner enclosure was added in order to better control mesh parameters near the rocket, with cushions of 5 D radial and 10 D frontal and rear, which was then extended to the rocket's wake. The lateral faces were made cylindrical to reduce volume and increase the uniformity of the boundary for better accuracy. The resulting enclosures are shown in Figure 3.6.5. Each body was assigned a random color automatically.

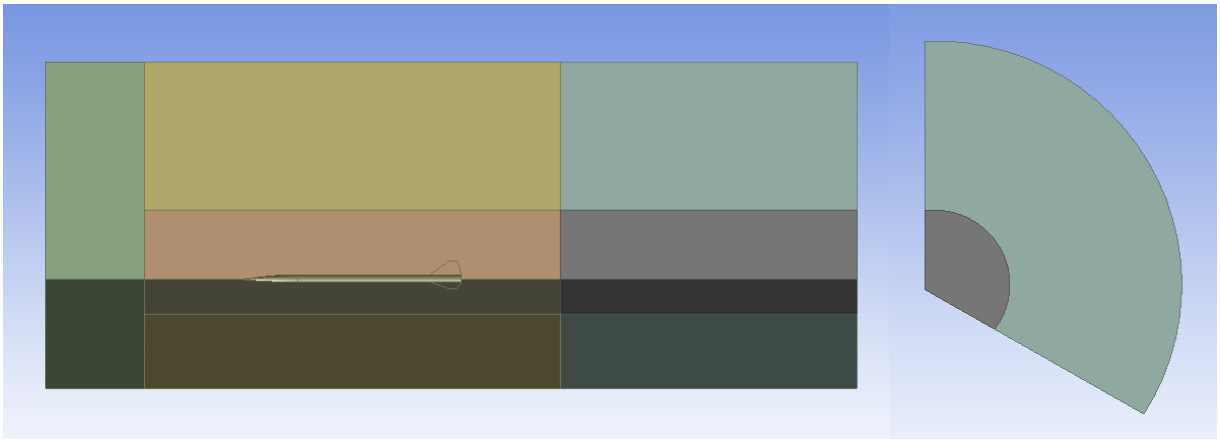


Figure 3.6.5: Enclosure boundaries

Next, meshing parameters were configured. Ideally, a k-omega SST tabulated turbulence model would be used to resolve the viscous sublayer in the boundary around the rocket. This model provides significantly higher drag accuracy, but takes significantly longer to compute and requires an extremely fine boundary layer, with a y+ value of 1 or lower. Using the standard Schlichting approximation for the skin friction coefficient [12], this would necessitate a

boundary first-layer thickness of 2.32 μm due to the high velocity. Therefore, it was decided that a standard wall function would be used, requiring a y+ between 30 and 300 for greatest accuracy [13], or a first-layer thickness between 0.07 and 0.7 mm. A y+ of 150 (first-layer thickness 0.35 mm) was used to balance accuracy and computation time, alongside the recommended minimum of 10 inflation layers with a standard growth rate of 1.2 [13]. The rocket body face sizing was held constant at 2.8 mm to reduce aspect ratio in the first layer and control boundary thickness. To preserve a smooth transition throughout the mesh, a global growth rate of 1.05 was imposed (and 1.01 in the innermost enclosures), and a full-tetrahedral patch conforming method was used. The mesh used for the 0% brake extrusion model is shown for example in Figures 3.6.6 and 3.6.7.

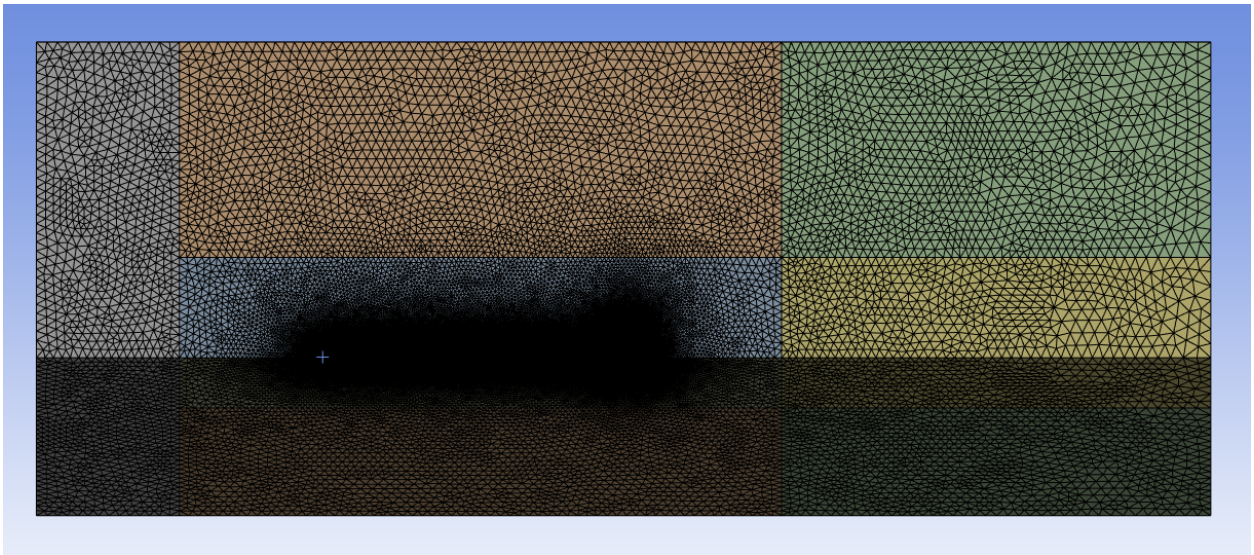


Figure 3.6.6: Global mesh

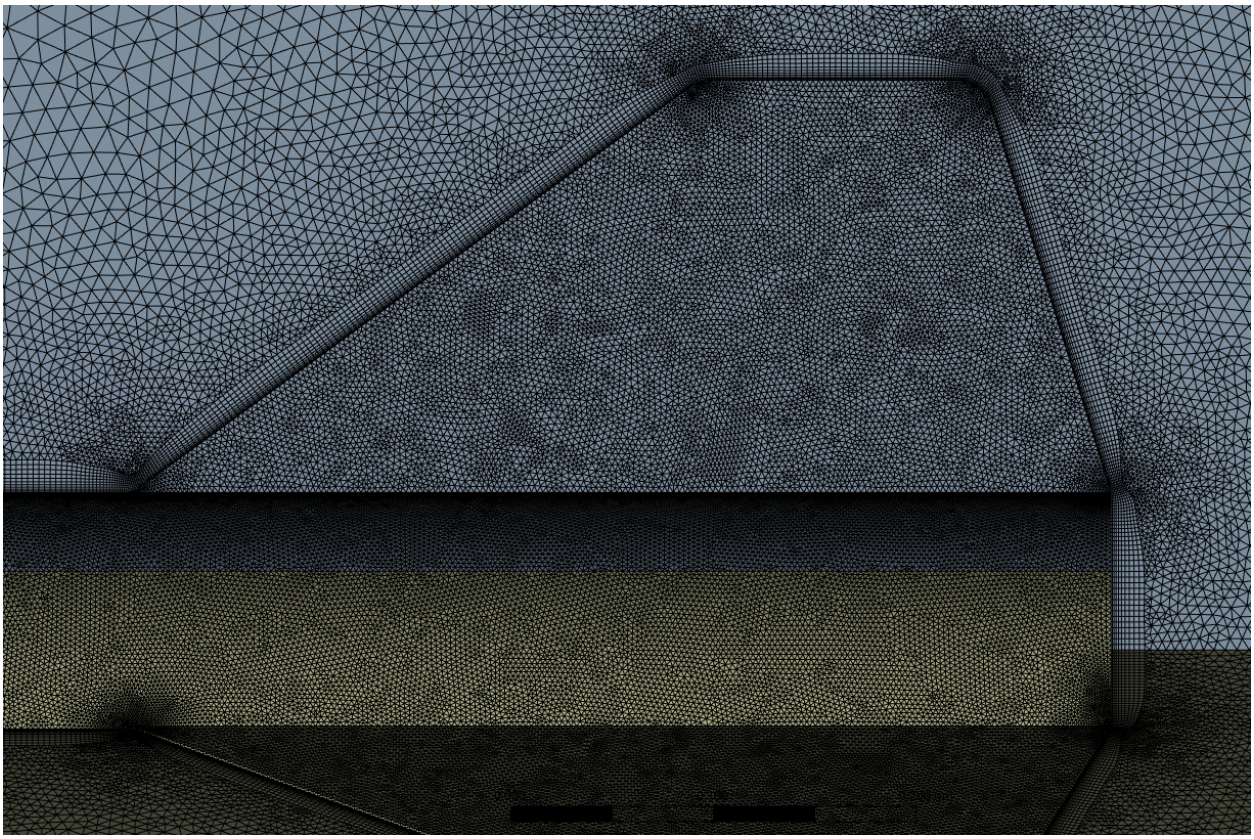


Figure 3.6.7: A segment of the inflation layers

This mesh consisted of 12.54 million elements, with a minimum orthogonality of 0.16. The maximum aspect ratio outside the boundary layer was 3.27, and inside the boundary layer ranged from 1.16 to 16.0. Maximum skewness was 0.85. The only significant difference in the 50% and 100% extrusion meshes was the element count, of 14 and 16 million respectively.

Finally, simulation parameters were configured. Due to the Mach number of 0.57, a density-based steady-state solver was used and the energy equation for an ideal gas was enabled. Standard sea level conditions were used. The velocity at the inlet was 195 m/s based on the most recent OpenRocket simulation data, the outlet was a pressure outlet, and the radial boundaries were pressure farfields. The rocket body was a stationary no-slip wall. Both axial slices were symmetry conditions with zero net mass flux. A k-epsilon turbulence model with compressibility effects and standard wall functions was used. Due to the high turbulence, particularly in the direct wake of the rocket, a 5-level FMG initialization with viscous terms was performed. Additionally, solution steering was enabled to dampen pseudo-transient flow features.

Convergence criteria were as follows:

- Flow calculation residuals below 1e-3
- Drag residual measured across 20 previous iterations, below 1e-3
- Inlet mass flow rate residual, measured across 20 previous iterations, below 1e-3

- Net mass flow rate error, normalized to inlet mass flow rate, below 1e-3

The drag results used in the flight simulations and controller are detailed in Table 3.6.1. The reference area for the drag coefficient is the cross-section of the rocket body, 0.019 m^2. The brake drag coefficient refers to the difference between the clean rocket drag and the other simulations' measured drag, to determine the effect of the brakes alone.

Table 3.6.1: CFD Drag Calculations

Deployment Level	Simulated Drag Value (N)	Calculated Drag Coefficient	Brake Drag Coefficient
0% (clean)	201	0.454	
50%	221.2	0.4997	0.0457
100%	270	0.6099	0.1559

3.7 Subscale Airframe Flight Test

Subscale flight testing was conducted on February 28, 2026, in accordance with launch site requirements, to validate recovery system functionality, stage separation, and overall structural integrity prior to full-scale flight operations. However, the airbrake system was operated in a data-collection-only mode, with deployment disabled. A smaller engine was used, based on the Tripoli requirement for the apogee of a rocket's first launch.

An M1186 motor was used, leading to an apogee of 4,202 ft. The primary objective of this test was to verify the recovery system's function. The vehicle showed a good ascent profile, characterized by a stable and vertical trajectory with no observable tilt or oscillatory behavior. All onboard avionics systems successfully powered on prior to launch, and the Featherweight GPS tracking system functioned throughout the entirety of the flight. The airbrake avionics system successfully recorded data.

At apogee, the recovery system experienced a partial failure in the primary deployment sequence. Although the Blue Raven altimeter registered the apogee event and indicated that a deployment signal was issued, the corresponding primary ejection charge (4 grams of black powder) was never ignited. As a result, stage separation and drogue deployment were not initiated by the primary system. The secondary EasyMini altimeter successfully detected apogee and initiated the backup ejection charge, successfully deploying the drogue parachute. The drogue parachute brought the terminal velocity to 116 ft/s.

A second failure occurred during the main parachute deployment phase. Neither the primary nor secondary system successfully deployed the main parachute. Post-flight analysis suggests that these failures were caused by the external toggle switch used to power the EasyMini system, as

the EasyMini lost power for several seconds during the time it should have deployed the main parachute. Despite passing pre-flight airflow testing, it is likely that vibration during decent caused the toggle switch to fail, resulting in loss of power to the secondary altimeter system.

The rocket impacted the ground at an estimated velocity of 116 ft/s, roughly five times faster than it was expected to descend under the main parachute. The nose cone cracked, the fins popped off the fin can, and the motor case dislodged itself from the inside of the fin can. However, the body tubes remained unharmed. The voltage regulator in the airbrake coupler and the PCB were lightly damaged, but still functioned. Both were replaced for the full-scale flights as a safety precaution. All other systems, including avionics and instrumentation modules, survived the impact.

Despite the recovery system issues, the flight test was considered partially successful. The rocket demonstrated phenomenal performance during ascent, and the secondary recovery system successfully compensated for primary system failure at apogee. Continuous GPS telemetry was maintained throughout the mission as well. Furthermore, the airbrake avionics systems successfully collected data, which were used to validate the accuracy of altitude and velocity estimation. Based on these results, the launch site granted approval for progression to full-scale flight testing.

After this flight, the use of externally mounted toggle switches for avionics activation was identified as an unnecessary risk. They were replaced with screw switches for subsequent tests. Additionally, the Blue Raven altimeter was removed, since it failed to deploy at all. The EasyMini was reconfigured as the primary altimeter, and a Missile Works RRC3+ was chosen as the replacement secondary due to its reputation as a robust system. Finally, the drogue shock cord was replaced due to visible fraying.

3.8 Full-Scale Base Flight Test

Full-scale airframe flight testing was conducted on April 11, 2026. This flight incorporated all of the design modifications implemented to address the recovery and avionics issues previously identified, as well as the rebuilt fin can and replaced nose cone.

The vehicle was flown using an M1655 motor. The airbrake system was activated for this flight but did not actuate the servo. This was later determined to be due to an issue with the IMUs used on the PCB, which had a default cap of 2 Gs, at which point they would cease to function.

The rocket exhibited a nominal ascent profile, maintaining a stable trajectory throughout the ascent with no observable deviation or oscillatory behavior. All onboard avionics systems performed as designed, with successful parachute deployment at the appropriate altitudes.

Post-flight inspection confirmed that the airframe sustained no structural damage. All onboard electronics were recovered in operational condition, with no evidence of power interruption or mechanical failure. The flight reached an apogee of 5867 ft.

3.9 Airbrake Flight Test

Due to the airbrakes not actuating on April 11, 2026, a follow-up launch took place on April 12, 2026, for 100% airbrake deployment. The purpose of this test was to determine the airbrakes' actual capability of apogee reduction, by comparison with the previous full-scale launch, discussed in section 3.8. Figure 2.6.4, included here for clarity, shows the airbrake system successfully demonstrated functionality during launch, achieving a 4.6% reduction in apogee from 5,867 ft to 5,597 ft.

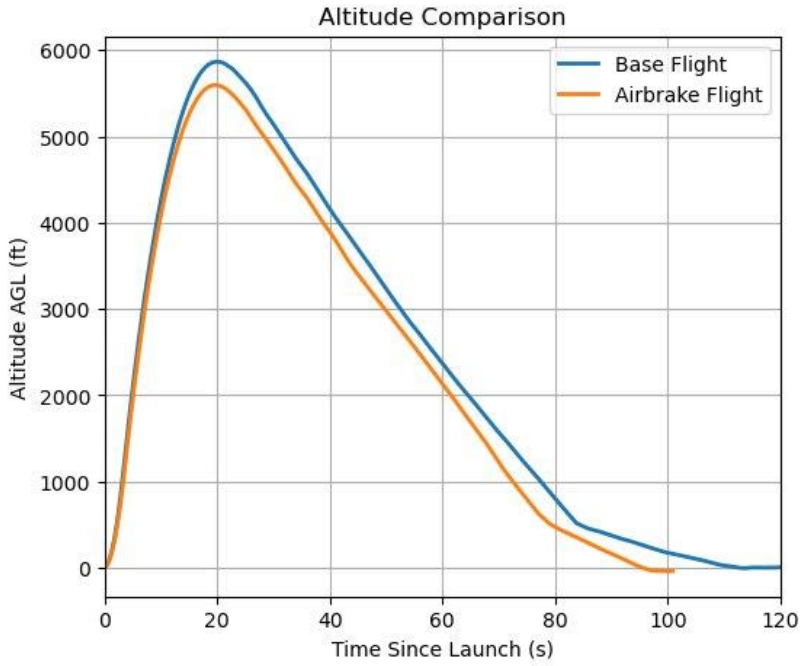


Figure 2.6.4: Flight Test Results

This confirms that the airbrakes work as intended.

4. Standards

This project must adhere to specific regulations in these four key areas: environmental regulations, FAA regulations, International Rocket Engineering Competition (IREC) requirements, and overall safety standards.

A. Environmental Regulations

The United States Environmental Protection Agency (EPA) defines standard to control hazardous emissions under the National Emission Standard for Hazardous Air Pollutants (NESHAP). These regulations apply to engine test cells that produce high levels of pollutants such as toluene, benzene, xylenes, and 1,3-butadiene [14]. The code of federal regulations Title 40, Part 61, Subpart D, also specifies additional standard that must be adhered to if the engine uses beryllium [15]. However, the engine developed for this project does not use any of these pollutants, confirming that from an environmental standpoint, the engine does not produce significant pollutants and remains compliant with the relevant standards.

B. FAA Regulations

The general operating limitations in 14 CFR 101.23 require that an amateur rocket be unmanned, suborbital, and not enter foreign airspace without prior approval. Additionally, 14 CFR 101.25 specifies that rockets using an L-class motor must follow several restrictions: the rocket may not launch within 1,500 ft of any person or property, nor within 5 nm of an airport without authorization [16]. Launches are also prohibited through clouds or when the projected apogee is under 50% cloud cover. Visibility must be at least 5 horizontal miles, and launches into controlled airspace require authorization.

Because the Tripoli Central Virginia BattlePark launch site lies within Class D controlled airspace, FAA authorization is required under 14 CFR 101.25(f). In accordance with 14 CFR 101.29(a), information such as the number of rockets, motor class, fuel and oxidizer type, launch site coordinates, expected apogee, and descriptions of the launch and recovery systems must be submitted to the FAA at least 45 days prior to launch via Form 7711-2.

In addition to FAA regulations, the Tripoli Central Virginia launch site requires compliance with the Tripoli Launch Safety Code [16], and NFPA 1127 [18]. Tripoli also requires submitting propellant details, including propellant name, mixture formula, weight (g), motor case size and designation, total impulse, maximum pressure, nozzle throat diameter, and burn-area-to-throat-area ratio two weeks before launch [19].

The rocket must adhere to all of these regulations and ensure that the necessary forms and documentation must be submitted on time.

C. IREC Requirements

There are several categories of regulations defined by IREC, including competition class whether it is 10,000ft, 30,000ft or 45,000 ft, propulsion rules, payload requirements, avionics and recovery requirements, and administrative compliance [20, 21, 22]. IREC requires every rocket to carry an active GPS tracker so that the vehicle can be located and recovered after flight, and the team must also demonstrate that their GPS tracker is functioning properly before launching. Additionally, every rocket must carry a commercial barometric altimeter to log altitude during flight, which must be analyzed and submitted for competition grading [20, 21].

Flight performance is the largest single scoring category, evaluating how accurately the rocket reaches its assigned target altitude and whether it is successfully recovered. For the rocket capstone team, hitting the target altitude is worth roughly 35% of the overall competition score, which emphasizes the importance of developing an altitude-targeting airbrake system.

D. Safety Standards

The primary physical safety code for high-power rocketry is NFPA 1127, which establishes the rules for launch-site conduct [23]. It specifies minimum safe distances, ignition-system requirements, recovery readiness procedures, and the authority of the Range Safety Officer (RSO). The RSO has full control over launch operations and may halt or deny any flight that poses a safety risk. NFPA 1127 also mandates strict arming procedures, including installing igniters last and performing continuity checks only when all personnel are at a safe distance [23]. Recovery systems must also be verified as fully prepared before the rocket leaves the preparation area. These rules are designed to prevent premature ignition and pad-level incidents. In addition, the Virginia BattlePark site requires teams to demonstrate the reliability of their recovery system through a ground test witnessed by a board member.

Overall, this project must comply with environmental regulations, FAA requirements, IREC competition rules, and safety standards. Ensuring full adherence through timely submission of documentation, and verification of electronics functionality will allow the team to have a smooth launch of the rocket.

5. Budget

The budget, including expenditures and sources of funding, is provided in Figures 5.1 and 5.2. The project had two primary sources of funding: the GW Rocket team's sponsorship (provided by the propulsion division) and the capstone budget provided by the university. The rocket team's budget paid for propellant mixing materials and recovery hardware, and the team provided spare aerostructure components at no cost to the capstone team. The capstone budget was primarily used for the airbrake mechanism, avionics, electronics, and control hardware.

Although there was a successful launch and demonstration of apogee reduction, the budget had a significant impact on testing, and therefore the development of the design, due to the propellant limit. Due to the high cost of manufacturing propellants, only a limited amount could be allocated for the capstone rocket launches. This included the subscale and two full-scale launches. Luckily, the GW Rocket team used the same subscale motor, so that static-fire test served a dual purpose.

Even with these limited amounts of propellant, the team was able to demonstrate successful launches but could not fully expand on the project. Additional budget would have allowed for more test launches to further refine the software and electronics of the airbrake avionics. Additional launches would allow for more testing and an iterative approach to design of the flight controller system, allowing the team to fully tune the controller over multiple flight attempts. The final airbrake launch was used primarily to demonstrate measurable apogee reduction and integrated system functionality.

Part Name	Quantity	Part Cost	Date Ordered	Date Recieved	Description	Company	Order Link	Quantity Cost	Estimated 10% shipping	Actual Cost	Order #
62kg Servo Motor	1	$89.99	10/21/2025	11/7/2025	Servo Motor for Air-brake deployment	Amazon	Order Link	$89.99	$98.99	$89.09	1
3S LiPo Battries	1	$74.98	10/21/2025	10/22/2025	Battery Pack with safe storage bag	Amazon	Order Link	$74.98	$82.48	$74.98	1
Teensy 4.1	1	$34.22	11/3/2025	11/7/2025	N/A	Amazon	Order Link	$34.22	$37.64	$35.25	2
MPU6050	1	$10.99	11/3/2025	11/7/2025	the 3 pieces one	Amazon	Order Link	$10.99	$12.09	$10.89	2
BMP388	1	$15.95	11/3/2025	11/6/2025	N/A	Amazon	Order Link	$15.95	$17.55	$14.99	2
Pen	1	$44.99	11/3/2025	11/6/2025	16 GB version, no Audio	Other	Order Link	$44.99	$49.49	$59.98	2
BNO080	1	$17.59	11/3/2025	11/7/2025	N/A	Amazon	Order Link	$17.59	$19.35	$17.49	2
BEC Voltage Regulator	1	$9.29	11/3/2025	11/7/2025	N/A	Amazon	Order Link	$9.29	$10.22	$9.29	2
SanDisk 128GB Ultra microSDXC	1	$13.99	11/3/2025	11/7/2025	128 GB x 1 pc	Amazon	Order Link	$13.99	$15.39	$14.79	2
Bearing	1	$3.90	11/13/2025	11/18/2025	1/2 hex by 1.125" OD flange bearing	Other	Order Link	$3.90	$4.29	$18.62	3
Countersunk Bolts	1	$5.99	11/13/2025	11/18/2025	M3-0.5 x 12mm countersunk bolt - pack of 40	Amazon	Order Link	$5.99	$6.59	$5.99	3
Threaded Rod	3	$4.45	11/13/2025	11/14/2025	1/4"-20 x 2.5 ft stainless steel	McMaster Carr	Order Link	$13.35	$14.69	$15.49	3
Hex Nuts	1	$2.49	11/13/2025	11/14/2025	1/4"-20 stainless - pack of 50	McMaster Carr	Order Link	$2.49	$2.74	$4.63	3
Primer	2	$12.95	11/13/2025	11/18/2025	N/A	Amazon	Order Link	$25.90	$28.49	$12.95	3
Cotton White Paint	2	$13.95	11/13/2025	11/18/2025	N/A	Amazon	Order Link	$27.90	$30.69	$13.95	3
Golden Elegance	1	$14.95	11/13/2025	11/18/2025	N/A	Amazon	Order Link	$14.95	$16.45	$14.95	3
Blue Like Denim	1	$13.95	11/13/2025	11/18/2025	N/A	Amazon	Order Link	$13.95	$15.35	$13.95	3
Same-Size Thread 18-8 Stainless Steel Shoulder Screw	1	$5.60	11/13/2025	11/14/2025	3 mm Shoulder Diameter, 12 mm Shoulder Length, M3 x 0.50 mm Thread, 4 mm thread length	McMaster Carr	Order Link	$5.60	$6.16	$7.74	3
Same-Size Thread 18-8 Stainless Steel Shoulder Screw	1	$5.25	11/13/2025	11/14/2025	3 mm Shoulder Diameter, 9 mm Shoulder Length, M3 x 0.50 mm Thread Size, 4 mm thread length	McMaster Carr	Order Link	$5.25	$5.78	$7.38	3
Same-Size Thread 18-8 Stainless Steel Shoulder Screw	1	$6.87	11/13/2025	11/14/2025	3 mm Shoulder Diameter, 14 mm Shoulder Length, M3 x 0.50 mm Thread	McMaster Carr	Order Link	$6.87	$7.56	$9.00	3
Loctite	1	$7.87	11/13/2025	11/18/2025	1 Pack	Amazon	Order Link	$7.87	$8.66	$7.87	3
Locknuts	1	$5.34	12/19/2025	1/12/2026	1/4"-20	McMaster Carr	Order Link	$5.34	$5.87	$12.69	4
Eye nut	1	$7.78	12/19/2025	1/12/2026	1/4"-20	McMaster Carr	Order Link	$7.78	$8.56	$15.13	4
Eye nut	2	$8.21	12/19/2025	1/12/2026	3/8"-16	McMaster Carr	Order Link	$16.42	$18.06	$23.77	4
Threaded rod	1	$7.20	12/19/2025	1/12/2026	1/4"-20 stainless	McMaster Carr	Order Link	$7.20	$7.92	$14.55	4
Pen	1	$69.99	12/19/2025	1/14/2026	16 GB version, with audio	Other	Order Link	$69.99	$76.99	$64.99	4
Adafruit 9-DOF Orientation IMU	1	$24.95	12/19/2025	1/12/2026	IMU Sensor	Other	Order Link	$24.95	$27.45	$35.52	4
Dry Lube	1	$7.35	12/19/2025	1/12/2026	Blaster Graphite 5.5 oz	Other	Order Link	$7.35	$8.09	$12.79	4
Bondo	1	$14.97	12/19/2025	1/12/2026	Short Strand Fiberglass Filler	Amazon	Order Link	$14.97	$16.47	$14.97	4
Toggle Switch	1	$6.39	1/12/2026	1/22/2026	On/Off toggle switches	Amazon	Order Link	$6.39	$7.03	$6.39	5
Threaded Rod	2	$4.45	1/12/2026	1/22/2026	1/4"-20 x 2.5 ft stainless steel	McMaster Carr	Order Link	$8.90	$9.79	$19.66	5
Step Down Converter	1	$12.99	1/26/2026	2/2/2026	DROK 5v Regulator, 5pcs Mini Voltage Reducer DC 4.5-24V	Amazon	Order Link	$12.99	$14.29	$12.96	6
Terminal Block	1	$5.99	1/26/2026	2/10/2026	Terminal Block,2 Pack 4 Circuits	Amazon	Order Link	$5.99	$6.59	$5.69	6
Wires	1	$15.69	1/26/2026	2/3/2026	40 ft 4 color electrical wire	Amazon	Order Link	$15.69	$17.26	$14.98	6
9 volt battery	1	$8.21	1/26/2026	2/2/2026	duracell 2 pack 9 volt battery	Amazon	Order Link	$8.21	$9.03	$8.21	6
Battery connector	1	$3.59	1/26/2026	2/2/2026	battery connector	Amazon	Order Link	$3.59	$3.95	$3.59	6
DROK DC Buck Converter	1	$15.99	1/26/2026	2/2/2026	Voltage Regulator 12V 6V-32V to 1.5-32V 5A	Amazon	Order Link	$15.99	$17.59	$15.96	6
30pcs 1/4-20 Hex Nut	1	$4.49	1/26/2026	2/2/2026	Hex Drive, UNC 1/4" Nuts Fasteners	Amazon	Order Link	$4.49	$4.94	$4.29	6
12 Pack Mini Lights	1	$8.99	2/3/2026	2/10/2026	12 Pack Mini Lights, Warm White Mini LED Spotlight	Amazon	Order Link	$8.99	$9.89	$8.99	7
LiPo Battery charger	1	$15.99	2/3/2026	2/10/2026	5 LiPo Batteries with a charger	Amazon	Order Link	$15.99	$17.59	$15.99	7
Threaded rod	5	$5.57	2/4/2026	2/10/2026	1/4"-20 stainless 3 ft long	McMaster Carr	Order Link	$27.85	$30.64	$52.24	8
DROK DC Buck Converter	1	$15.99	3/13/2026	3/25/2026	Voltage Regulator 12V 6V-32V to 1.5-32V 5A	Amazon	Order Link	$15.99	$17.59	$15.96	9
EasyMini	1	$80.00	3/13/2026	3/30/2026	EasyMini Barometer	Other	Order Link	$80.00	$88.00	$94.50	9
Missile Works Altimeter	1	$74.95	3/16/2026		RRC3+ Altimeter	Other	Order Link	$74.95	$82.45		10
Proline Epoxy	1	$39.00	4/1/2026		Proline Epoxy 4500P	Other	Order Link	$39.00	$42.90		11
O-rings	2	$11.71	4/1/2026		338 O-ring	McMaster Carr	Order Link	$23.42	$25.76		11
E-matches	1	$16.00	4/1/2026		Bundle of 20	Other	Order Link	$16.00	$17.60		11
									Total Spent so far	$ 933.08	
									Remaining $$ in Budget	$ 266.92	

Figure 5.1: Capstone Budget

Order Type	Part Name	Quantity	Part Cost	Date Ordered	Date Arrived	Description	Company	Order Status	Order Link	Notes	Quantity Cost	Cost with Estimated 10% shipping
Motors	Subscale Equipment	1	$ 575.00			Subscale Equipment	Amazon	Not Ordered	https://binderdesign.com	N/A	$ 575.00	$ 632.50
Recovery Hardware	Main Parachute	1	$ 548.88	11/2/2025	1/12/2026	Fruity Chutes Iris Ultra Compact 96"	Fruity Chutes	Delivered	https://shop.fruitychutes.	Orange/Black Parachute	$ 548.88	$ 603.77
Recovery Hardware	Drogue Parachute	1	$ 81.59	11/2/2025	11/30/2025	Fruity Chutes 18" Elliptical Drogue	Fruity Chutes	Delivered	https://shop.fruitychutes.	Stock Color	$ 81.59	$ 89.75
Recovery Hardware	Drogue Shock Cords	2	$ 63.00	11/2/2025	11/30/2025	40 ft	Madcow	Delivered	https://www.madcowrocl	N/A	$ 126.00	$ 138.60
Recovery Hardware	Main Shock Cords	2	$ 43.00	11/2/2025	11/30/2025	30 ft	Madcow	Delivered	https://www.madcowrocl	N/A	$ 86.00	$ 94.60
Recovery Hardware	Main Deployment Bag	1	$ 50.50	11/2/2025	11/30/2025	5" x 10" main deployment bag	Rocketman	Delivered	https://the-rocketman.co	Rocketman Deployment I	$ 50.50	$ 55.55
Recovery Hardware	Drogue Deployment Bag	1	$ 28.50	11/2/2025	11/30/2025	2" x 8" drogue deployment bag	Rocketman	Delivered	https://the-rocketman.co	Rocketman Deployment I	$ 28.50	$ 31.35
Recovery Hardware	Coupler Bulkplates	6	$ 9.90	11/2/2025	11/14/2025	6 inch diameter	Other	Delivered	https://wildmanrocketry.c	n/a	$ 59.40	$ 65.34
Recovery Hardware	Featherweight GPS	1	$ 380.00	11/2/2025	11/30/2025	Full system w/antennas	Featherweight	Delivered	https://www.featherweigl	Full System with wire whi	$ 380.00	$ 418.00
Recovery Hardware	Featherweight Altimeter	1	$ 175.00	11/2/2025		Featherweight altimeter	Featherweight	Delivered	https://www.featherweigl	One Blue Raven	$ 175.00	$ 192.50
Recovery Hardware	EasyMini Altimeter	1	$ 111.96	11/2/2025	11/14/2025	EasyMini altimeter	Apogee	Delivered	https://www.apogeerocke	N/A	$ 111.96	$ 123.16
Recovery Hardware	Swivel	4	$ 16.90	11/2/2025	11/14/2025	5/16" swivel	McMaster Carr	Delivered	https://www.mcmaster.c	N/A	$ 67.60	$ 74.36
Recovery Hardware	Quick link	8	$ 5.91	11/2/2025	11/14/2025	9/32 stainless	McMaster Carr	Delivered	https://www.mcmaster.c	N/A	$ 47.28	$ 52.01
Recovery Hardware	Schuetzen black powder	1	$ 21.30			blackpowder	Other	Not Ordered	https://www.buffaloarms	N/A	$ 21.30	$ 23.43
Other	Aluminum	1	$ 26.57			Spare Aluminum	Amazon	Not Ordered	https://www.amazon.com	N/A	$ 26.57	$ 29.23
Recovery Hardware	Nylon Shear Pins	2	$ 7.96	11/2/2025	11/14/2025	Large Nylon Pins	Apogee	Delivered	https://www.apogeerocke	N/A	$ 15.92	$ 17.51
Other	Rail Buttons				1/12/2026							
											Total w/out Shipping	Total w/ Shipping
											$ 2,401.50	$ 2,641.65
										Ordered through rocket	$1946.59 after tax	

Figure 5.2: Rocket-Provided Capstone Budget

6. Summary and Recommendations

The development and testing of the altitude targeting rocket system showed that the overall design is structurally sound, stable in flight, and capable of supporting all required subsystems including propulsion, recovery, and the airbrake system. Across both subscale and full-scale launches, the rocket consistently flew straight with no noticeable tilt angles or oscillations. Although the rocket suffered damage during subscale testing, the rocket was able to be rebuilt and tested again, highlighting the work done to identify and mitigate all points of failure.

The system meets all basic requirements. The airbrake system fits within the required volume constraints and was successfully integrated into the rocket. During flight, the airbrakes were able to achieve a 4.6% reduction in apogee, satisfying the need to be able to produce apogee reduction as well as withstand the resultant drag force on the system. In addition, the airbrake system was within the weight requirement specified in Table 1.3.1. Finally, the rocket was also able to safely land and be reused, meeting all requested functional requirements.

The rocket stayed well within the budget of the project. A minimum apogee of 4500 ft was achieved during full-scale flight testing. On-board diagnostic and cinematic cameras were integrated into the rocket. The controller software was developed and simulated but was not able to be tested in-flight. In addition, a self-defined desirable apogee reduction of 7.5% was not achieved.

If the GW Rocket team chooses to implement this design for future rocket launches, the system should continue to be developed and refined, as it has been shown to be a useful approach for altitude targeting. The main aspect left to be tested is the feedback control loop, which was developed and simulated but not validated in flight. If successful, implementation of this system would result in GW being the first university internationally to fly a successful airbrake system at IREC.

7. Acknowledgements

The authors would like to thank Professor Amir Barakati, Professor Steven Shooter, and Professor Mitchell Narins for their mentorship, advice, and aid throughout the project. The authors would also like to thank Professor Elias Balaras, Professor Azar Panah, Professor Kartik V. Bulusu, and Professor Michael W. Plesniak for listening to our inquiries about CFD and GW-owned wind tunnel accessibility throughout the semester. The authors would also like to thank Bill Rutkowski, Nick Batista, and Tom Punte from the machine shop for their aid in manufacturing. Additionally, we would like to thank Ibrahim Yildiran for all his help in teaching the team how to approach CFD simulations, as well as Joe Creech from GW's Research and Technology Services office for providing the team with a virtual desktop interface to run CFD simulations, which made all the difference in the end.

The authors particularly wish to acknowledge Ben and Elaine Russell for their advice and mentorship in the propulsion and avionics aspects of the project. Their experience and ability to address any rocket-related questions the team had helped us stay on track and avoid wasting time on bad decisions.

8. References

[1] Farha, Michael L., 2025, "Influence of Actuated Air Brakes on the Apogee of Sub-Orbital Rockets," Graduate Theses, Dissertations, and Problem Reports, https://researchrepository.wvu.edu/etd/12808

[2] Sprague, T., Williams, J., & Smith, C. (2024, April). Design and analysis of a high-powered rocket airbrake system. Paper presented at the AIAA Regional Student Conference, Starkville, MS.

[3] Sanipara, N., Belew, A., and Sprague, T., 2024, "Design and Analysis of a High-Powered Rocket Airbrake System," Regional Student Conferences, American Institute of Aeronautics and Astronautics, 2024, pp. 1–11. https://doi.org/10.2514/6.2024-85628.

[4] Sengupta, A., 2025, "Rocket Airbrakes," *Armaan Sengupta*. https://www.armaansengupta.ca/rocketry.

[5] RocketPy Contributors, "Airbrakes Module," RocketPy Documentation, https://docs.rocketpy.org/en/latest/user/airbrakes.html, 2025.

[6] ThrustMIT, "Rocket Airbrakes and Flight Modeling," Peak of Flight Newsletter, Issue 599, Apogee Rockets, 2024, https://www.apogeerockets.com/Peak-of-Flight/Newsletter599.

[7] Leroy Musa, The Arbalest Goose 4 Avionics Systems & Flight Computer Software, 2024, https://github.com/Arbalest-Rocketry/flight-computer/blob/master/README.md

[8] Adafruit Blog, Small Form Factor Amateur Rocket Avionics and Flight Software, 2025, https://blog.adafruit.com/2025/02/11/small-form-factor-amateur-rocket-avionics-and-flight-software/

[9] NASA, Flight Software & Embedded System Framework, 2020, https://nasa.github.io/fprime/

[10] Morris S. Berman, Electronic Components for High-g Hardened Packaging, Army Research Laboratory, 2007, https://apps.dtic.mil/sti/pdfs/ADA443252.pdf

[11] "Rocket Weathercocking," NASA, https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/rocket-weather-cocking/.

[12] CFD Online, “Y plus wall distance estimation,” https://www.cfd-online.com/Wiki/Y_plus_wall_distance_estimation.

[13] Ansys, Inc., “Ansys Fluent User’s Guide,” 2023 R2 ed., 2023.

[14] Federal Register / Vol. 68, No. 101 / Tuesday, May 27, 2003 / Rules and Regulations https://www.govinfo.gov/content/pkg/FR-2003-05-27/pdf/03-5521.pdf

[15] National Emission Standard for Beryllium Rocket Motor Firing https://www.ecfr.gov/current/title-40/chapter-I/subchapter-C/part-61/subpart-D

[16] 14 C.F.R. § 101 (2025). https://www.ecfr.gov/current/title-14/chapter-I/subchapter-F/part-101

[17] Tripoli Rocketry Association. (2025, March). Safety Code. Tripoli Rocketry Association, Inc. https://www.tripoli.org/safetycode

[18] National Fire Protection Administration. (2025). NFPA 1127: 2026 Code for High Power Rocketry.

[19] Flight Requirements. Tripoli Central Virginia. (2025). https://battlepark.org/?page_id=1010

[20] Experimental Sounding Rocket Association (ESRA), International Rocket Engineering Competition (IREC) Rules & Requirements Document, Version 1.6, ESRA, 2025. https://www.soundingrocket.org/uploads/9/0/6/4/9064598/irec_rules_and_requirements_docume nt_v_1.6_.pdf.

[21] Experimental Sounding Rocket Association (ESRA), International Rocket Engineering Competition Design, Test, & Evaluation Guide (DTEG), Revision 2025 V1.1.6, ESRA, 2025. https://www.soundingrocket.org/uploads/9/0/6/4/9064598/2025_irec_dteg_v1.1.3_10-22-24.pdf

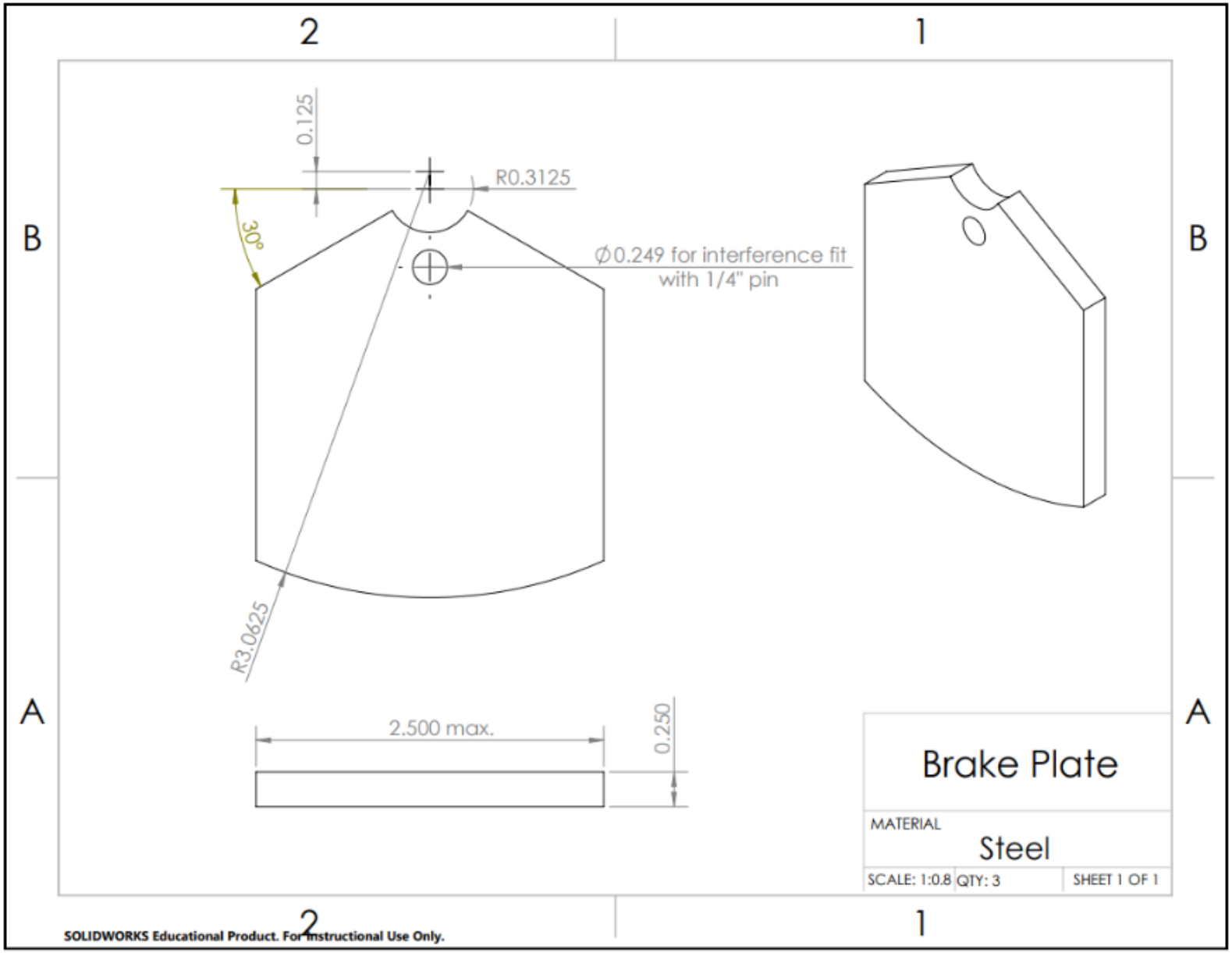
[22] Experimental Sounding Rocket Association (ESRA), “IREC Documents & Forms / Competition Resources,” ESRA, 2025 https://www.soundingrocket.org/irec-documents--forms.html.

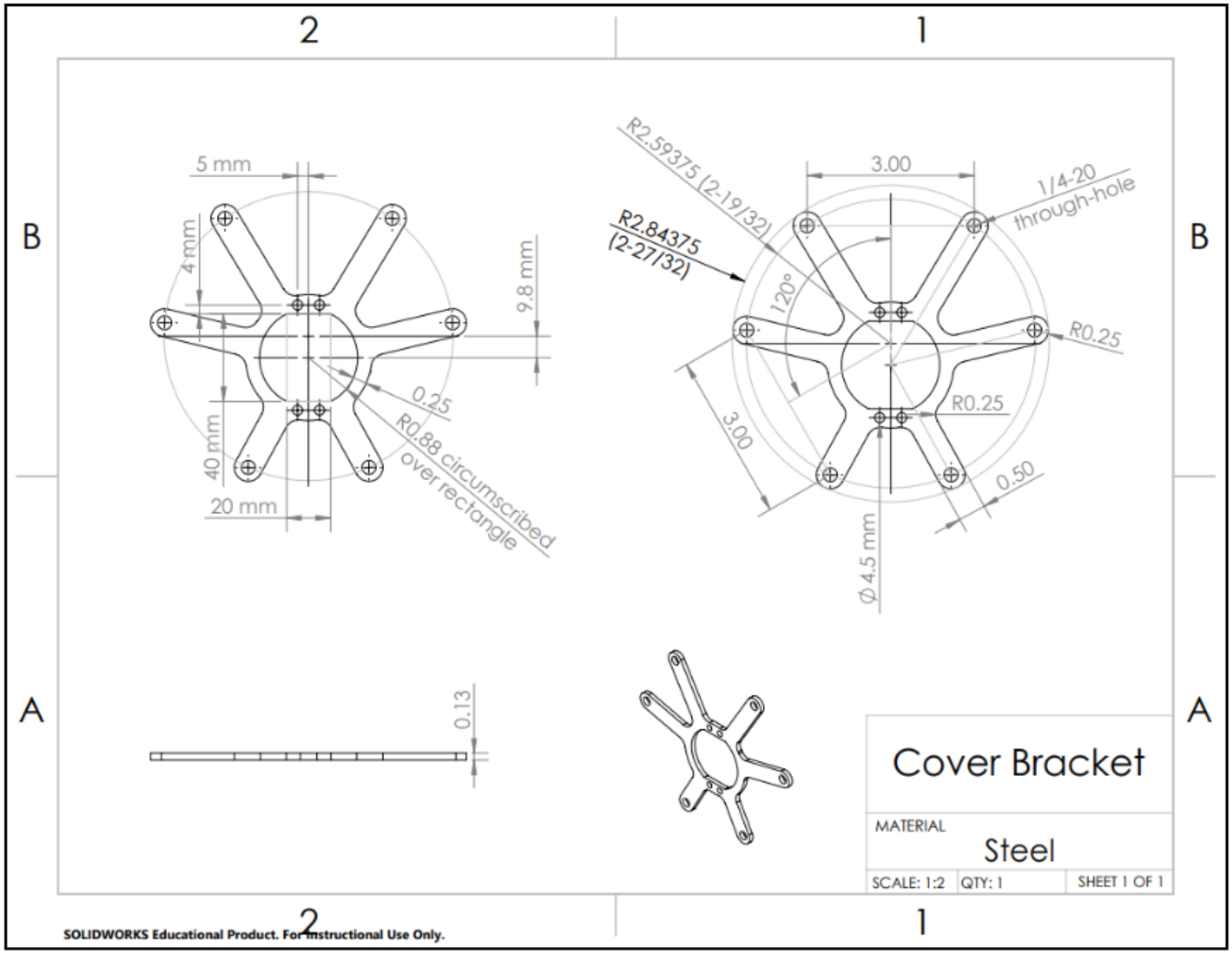
[23] National Fire Protection Association. (2008). NFPA 1127: Code for high power

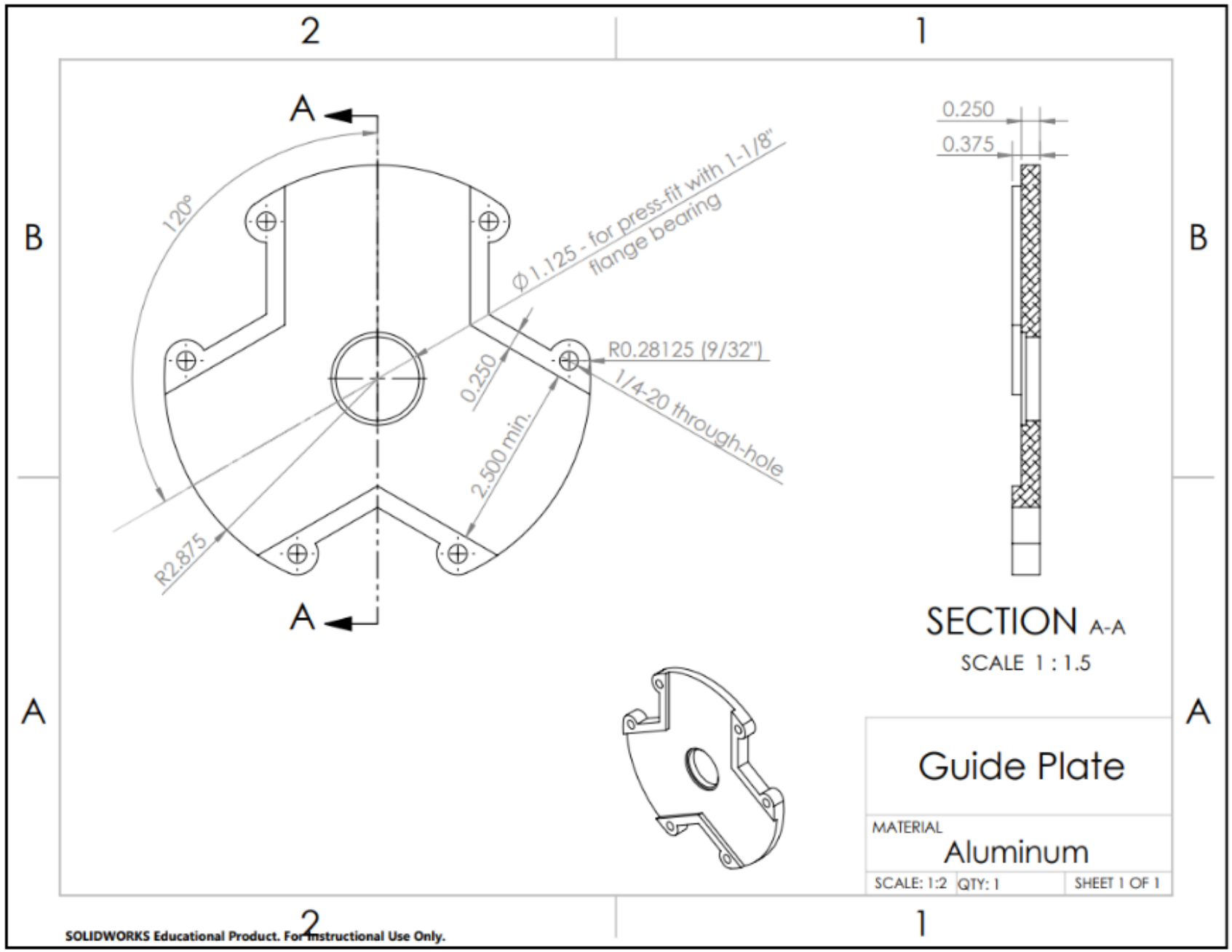
rocketry (2008 ed.). https://www.theweeks.org/tmp/FILES/NFPA-1127.pdf

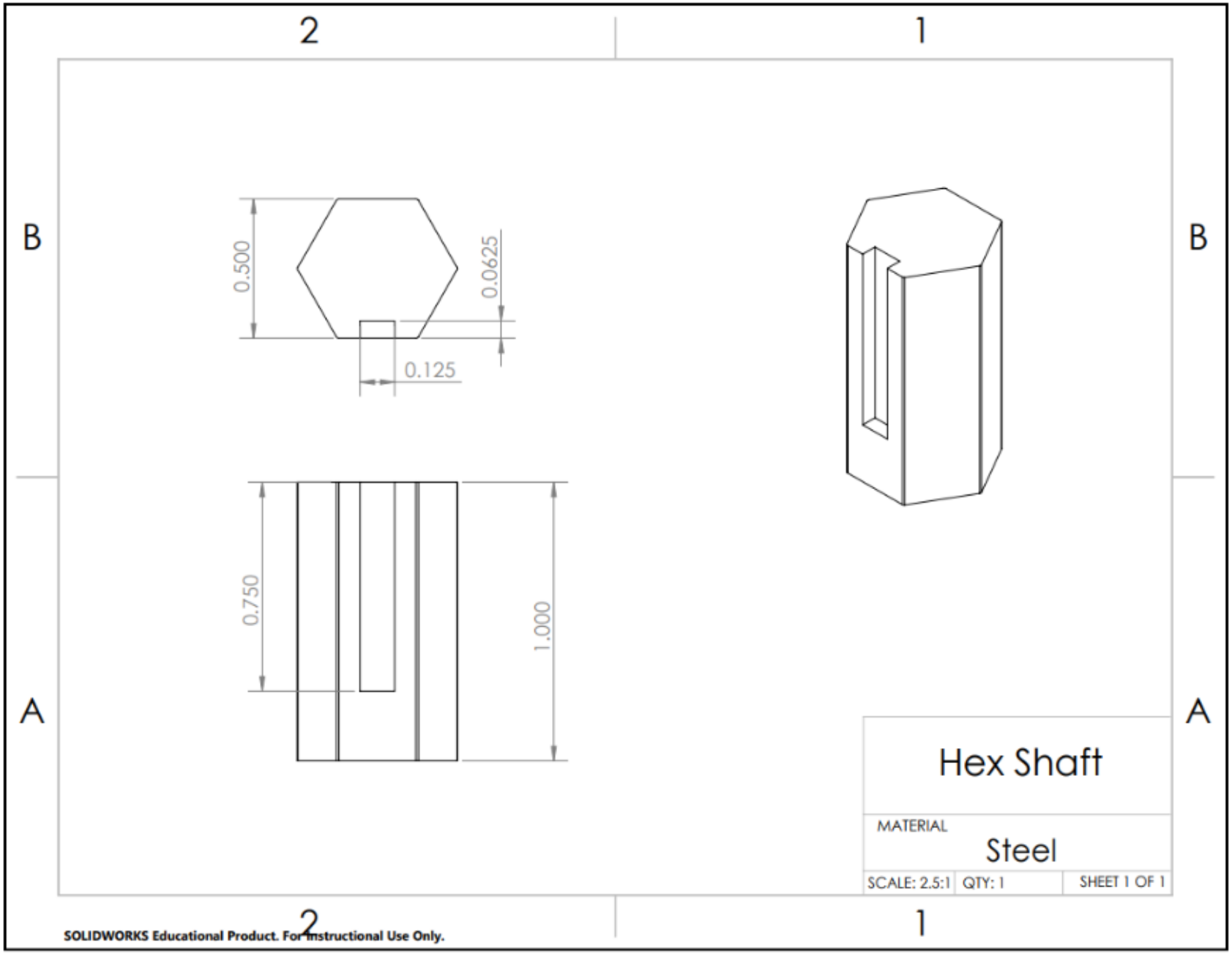
9. Appendices

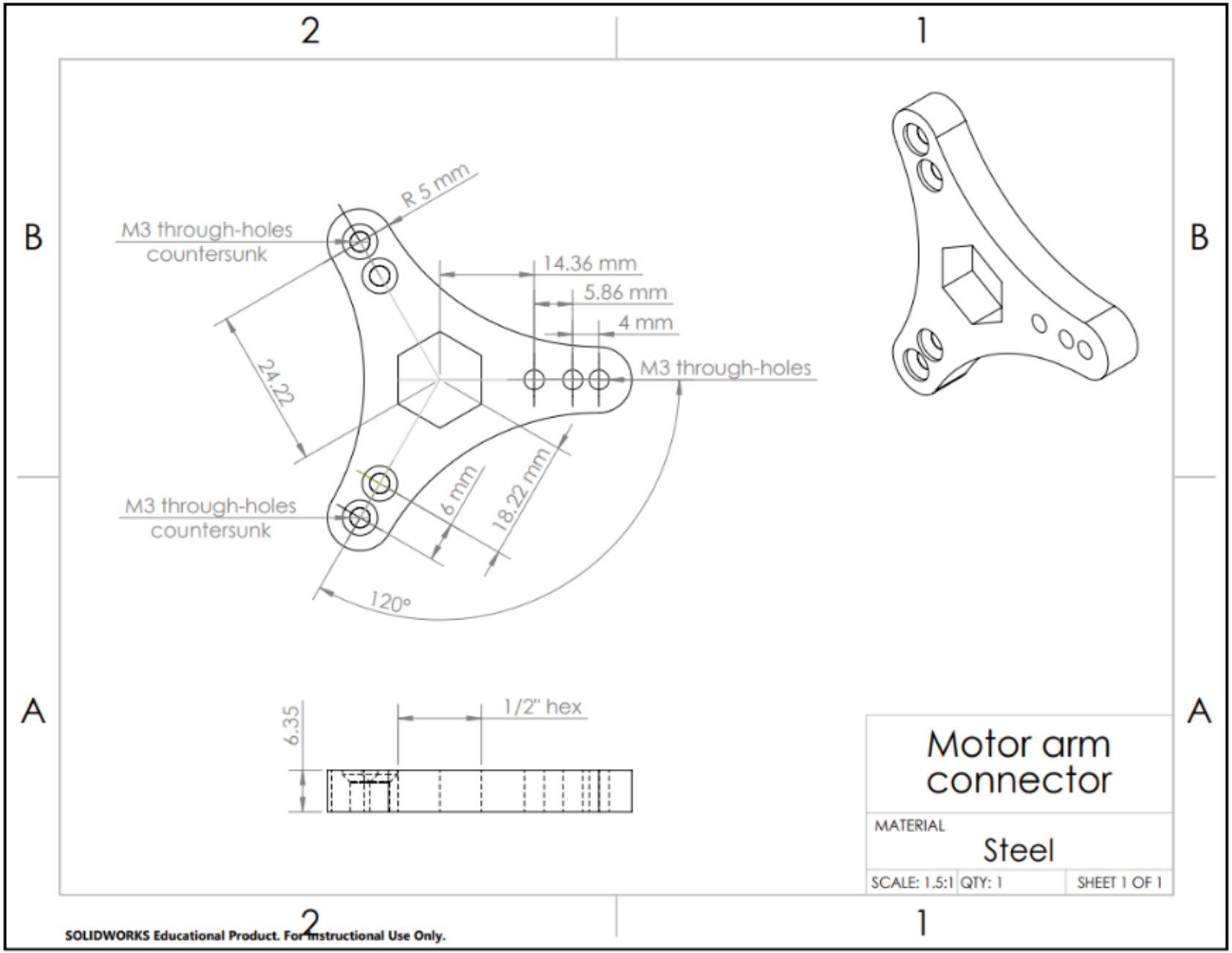
9.1 Manufacturing Drawings

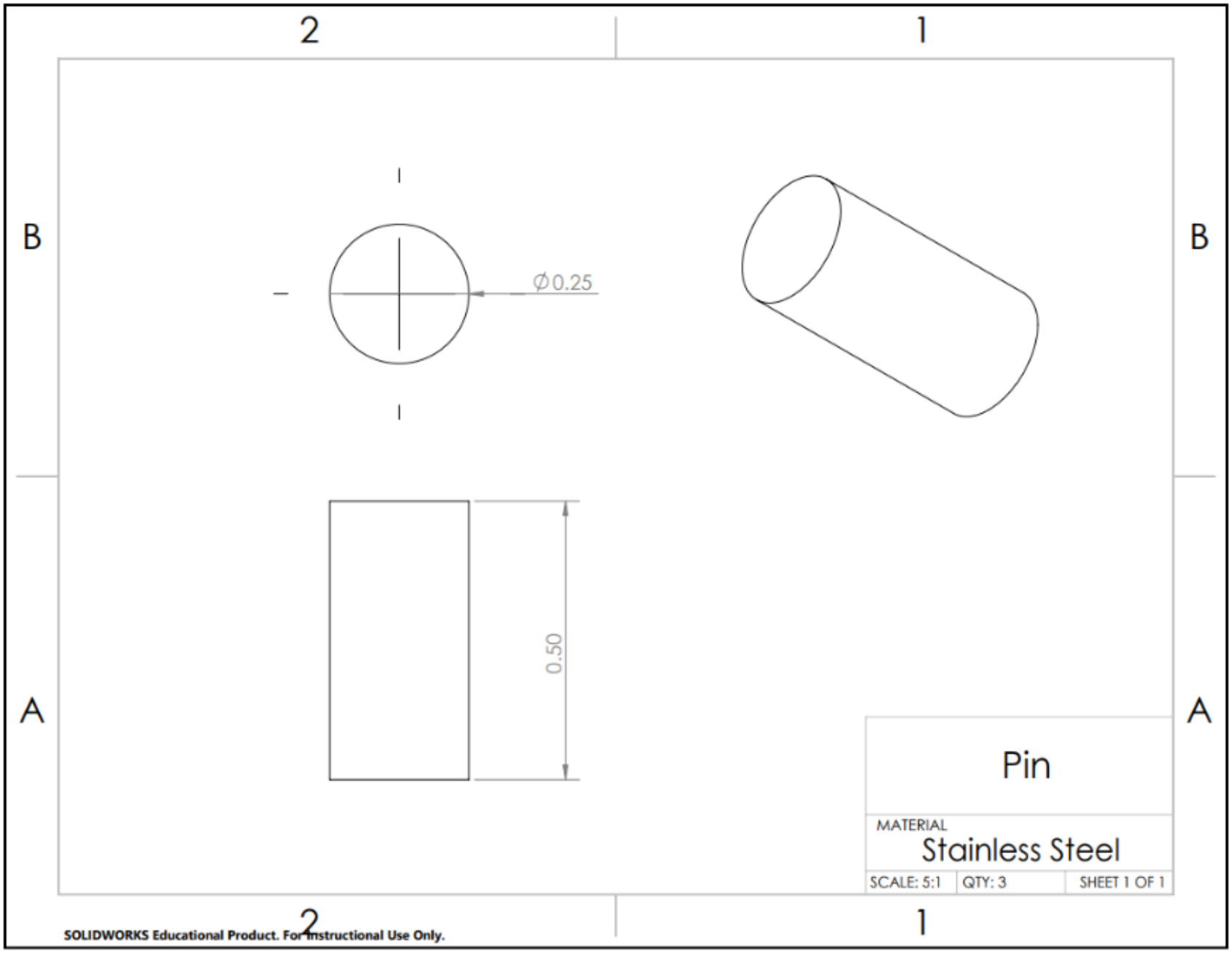

2
1
B
A
0.125
R0.3125
30°
Ø0.249 for interference fit
with 1/4" pin
R3.0625
2.500 max.
0.250
Brake Plate
MATERIAL
Steel
SCALE: 1:0.8
QTY: 3
SHEET 1 OF 1
SOLIDWORKS Educational Product. For Instructional Use Only.

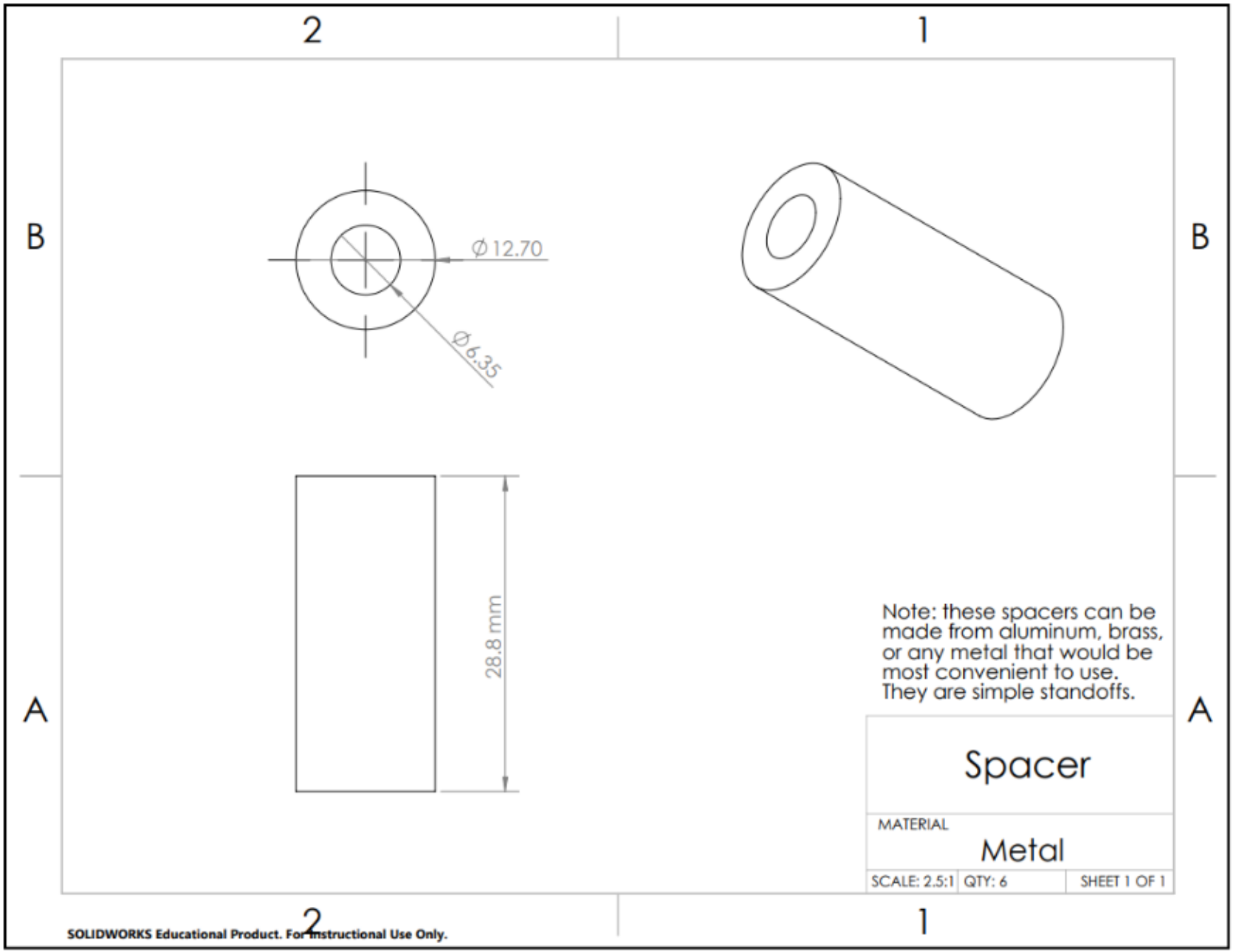

2
1
B
A
5 mm
4 mm
9.8 mm
40 mm
20 mm
0.25
R0.88 circumscribed over rectangle
0.13
R2.59375 (2-19/32)
R2.84375 (2-27/32)
3.00
1/4-20 through-hole
120°
R0.25
R0.25
3.00
0.50
Ø 4.5 mm
Cover Bracket
MATERIAL
Steel
SCALE: 1:2
QTY: 1
SHEET 1 OF 1
SOLIDWORKS Educational Product. For Instructional Use Only.

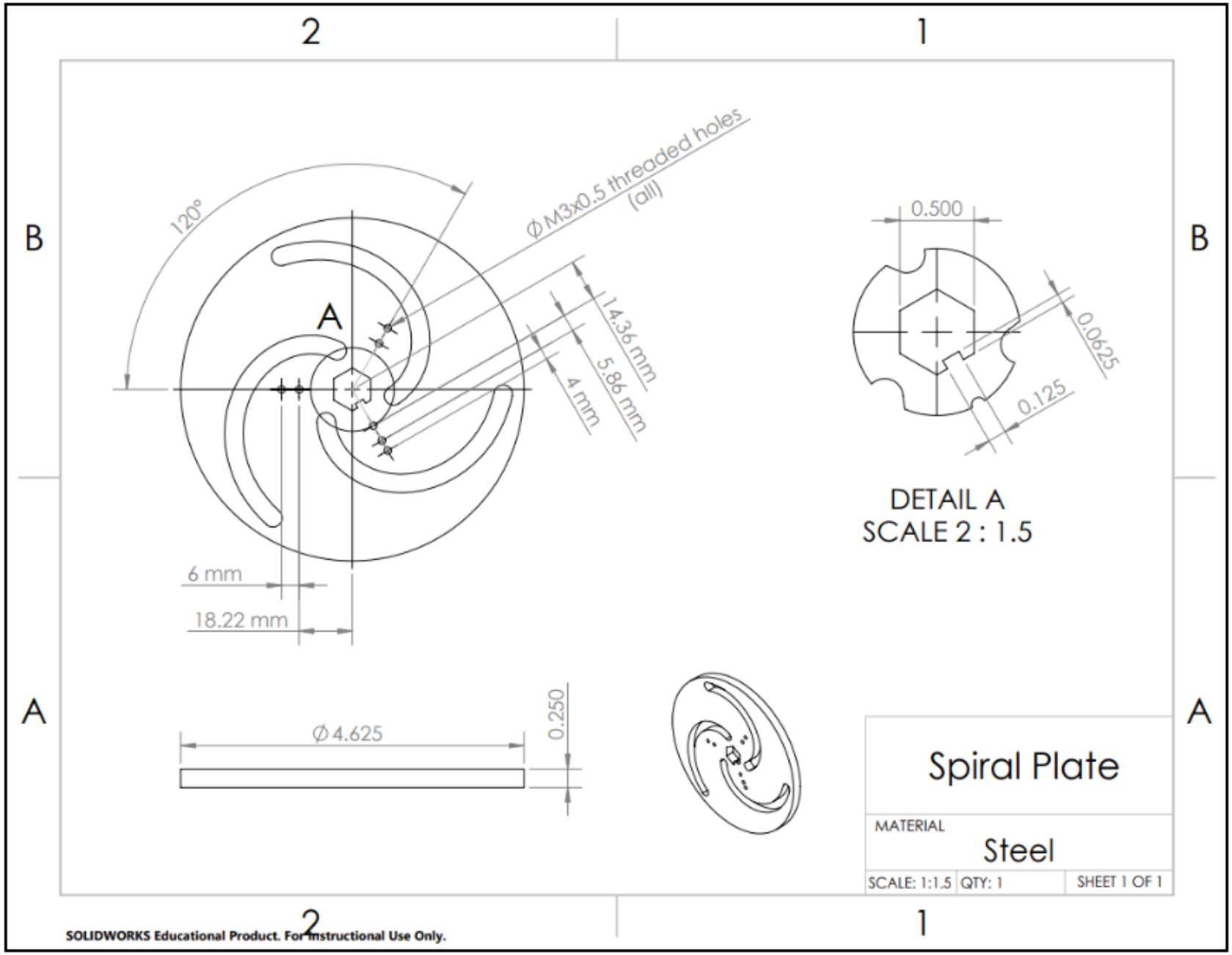

2
1
A
B
A
120°
Φ1.125 - for press-fit with 1-1/8" flange bearing
R0.28125 (9/32")
1/4-20 through-hole
0.250
2.500 min.
R2.875
A
0.250
0.375
SECTION A-A
SCALE 1 : 1.5
Guide Plate
MATERIAL
Aluminum
SCALE: 1:2
QTY: 1
SHEET 1 OF 1
SOLIDWORKS Educational Product. For Instructional Use Only.

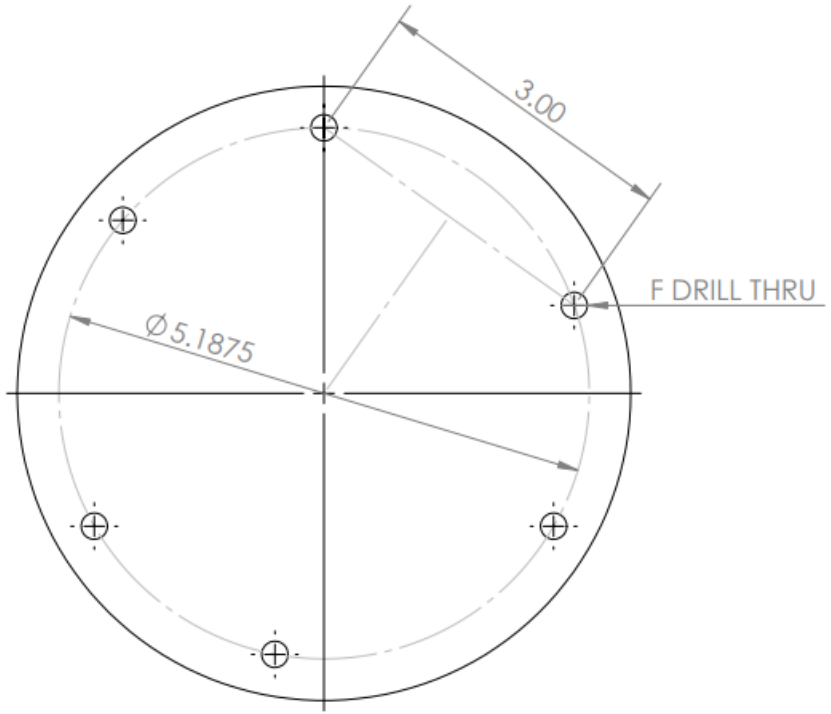

2
1
B
A
0.500
0.0625
0.125
0.750
1.000
Hex Shaft
MATERIAL
Steel
SCALE: 2.5:1
QTY: 1
SHEET 1 OF 1
SOLIDWORKS Educational Product. For Instructional Use Only.

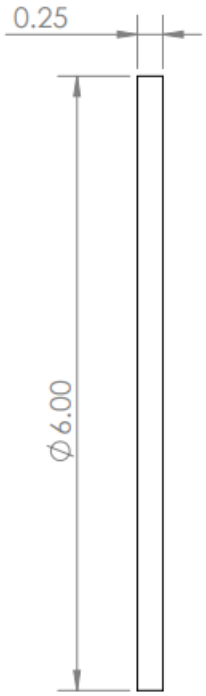

2
1
B
A
R 5 mm
M3 through-holes
countersunk
14.36 mm
5.86 mm
4 mm
M3 through-holes
24.22
M3 through-holes
countersunk
6 mm
18.22 mm
120°
6.35
1/2" hex
Motor arm
connector
MATERIAL
Steel
SCALE: 1.5:1
QTY: 1
SHEET 1 OF 1
SOLIDWORKS Educational Product. For Instructional Use Only.

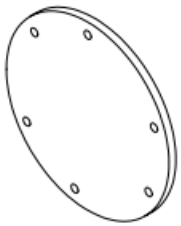

2
1
B
B
Ø0.25
0.50
A
A
Pin
MATERIAL
Stainless Steel
SCALE: 5:1
QTY: 3
SHEET 1 OF 1
2
1
SOLIDWORKS Educational Product. For Instructional Use Only.


2
1
B
B
A
A
Ø 12.70
Ø 6.35
28.8 mm
Note: these spacers can be made from aluminum, brass, or any metal that would be most convenient to use. They are simple standoffs.
Spacer
MATERIAL
Metal
SCALE: 2.5:1
QTY: 6
SHEET 1 OF 1
2
1
SOLIDWORKS Educational Product. For Instructional Use Only.


2
1
B
A
120°
Ø M3x0.5 threaded holes (all)
14.36 mm
5.86 mm
4 mm
6 mm
18.22 mm
Ø 4.625
0.250
0.500
0.0625
0.125
DETAIL A
SCALE 2 : 1.5
Spiral Plate
MATERIAL
Steel
SCALE: 1:1.5
QTY: 1
SHEET 1 OF 1
SOLIDWORKS Educational Product. For Instructional Use Only.





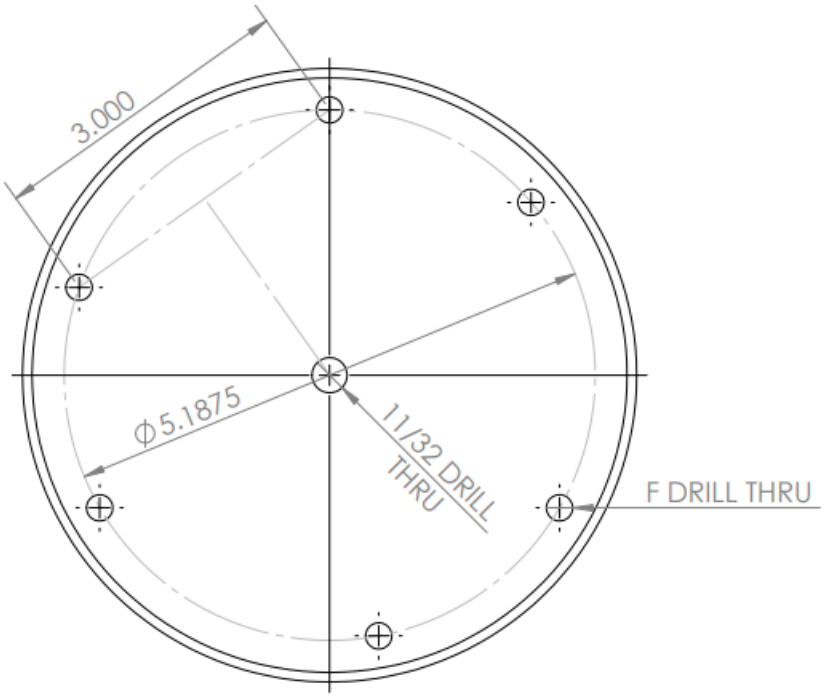


Airbrake Coupler Aft Bulkplate

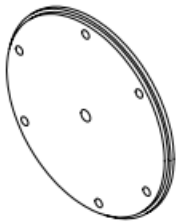
MATERIAL: Steel

SCALE: 1:0.8 | QTY: 1 | SHEET 1 OF 1

SOLIDWORKS Educational Product. For Instructional Use Only.





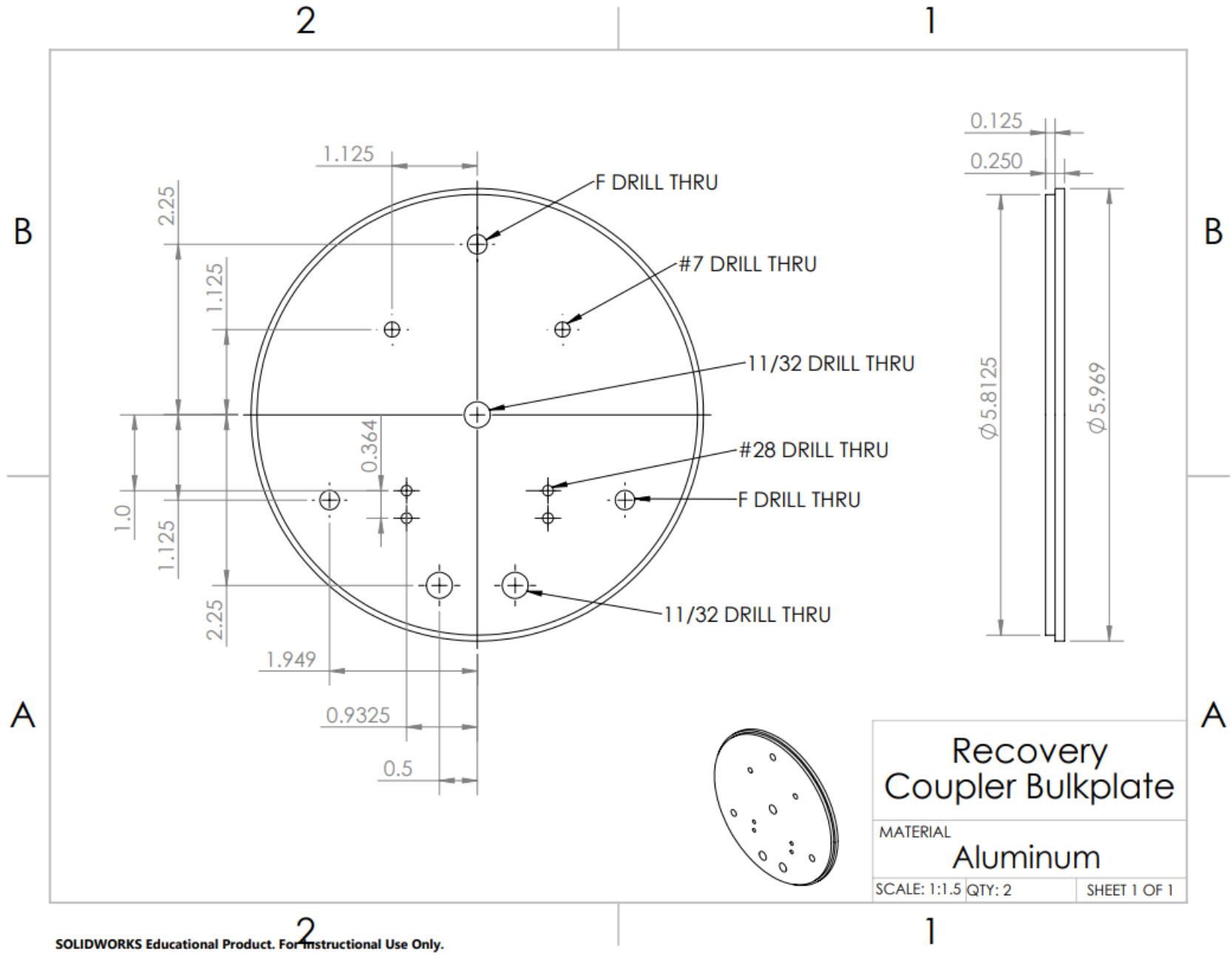


Airbrake Coupler Fore Bulkplate

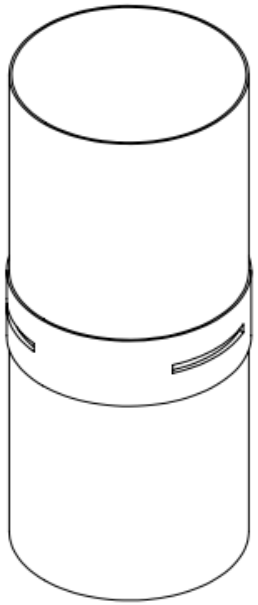
MATERIAL
Aluminum

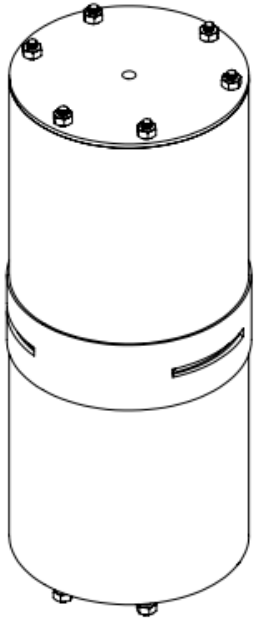
SCALE: 1:1.5 | QTY: 1 | SHEET 1 OF 1

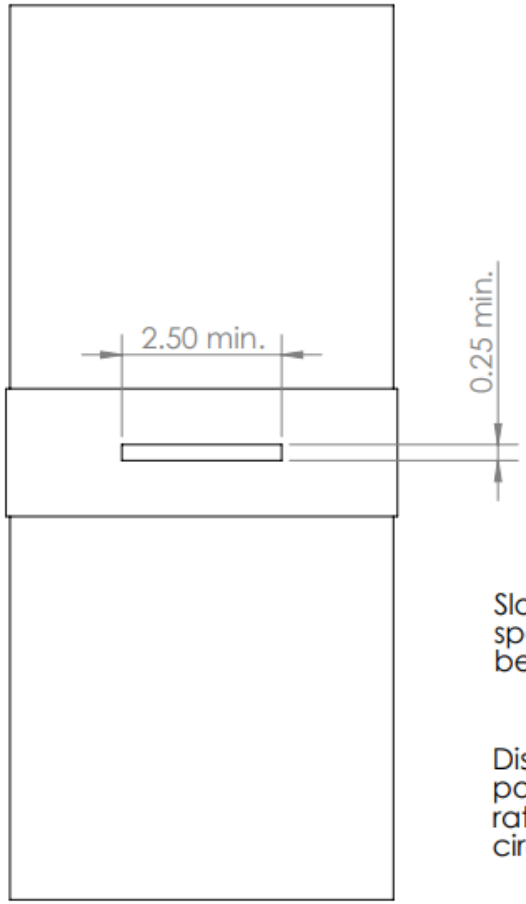
SOLIDWORKS Educational Product. For Instructional Use Only.



SOLIDWORKS Educational Product. For Instructional Use Only.







Slots should be spaced evenly between rivets

Distance is measured point-to-point, rather than along circumference

SOLIDWORKS Educational Product. For Instructional Use Only.

9.2 Launch Day Operations Flowchart

